



Department of Electronic & Telecommunication Engineering  
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EN4431 - ANALOG IC DESIGN

A PFD-CP TYPE-II  $\Delta\Sigma$  FRACTIONAL-N PHASE LOCKED LOOP SIMULATION

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## ABSTRACT

This report presents the design and simulation of a Type-II Phase-Frequency Detector and Charge Pump (PFD-CP) based Fractional-N Phase-Locked Loop (PLL) implemented in 45nm CMOS technology. The primary objective is to design a robust and low-power frequency synthesizer suitable for high-speed system-on-chip (SoC) applications requiring precise clock generation. The proposed architecture integrates a tri-state PFD, charge pump, loop-pass filter, ring voltage-controlled oscillator (VCO), and a 4-bit binary frequency divider in place of a delta-sigma modulator (DSM) to achieve fractional division without complexity. The design methodology includes detailed transistor-level sizing, verification through transient simulations, and FFT analysis to evaluate the output frequency spectrum and locking performance. Key performance metrics such as lock time, frequency stability, and power consumption are analyzed. Additionally, the report discusses an architectural modification to realize an Integer-N PLL by removing the DSM. Simulation results validate the design's capability to achieve a wide frequency range and fast locking with minimal power consumption, making it suitable for advanced AMS-PLL implementations.

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# 1 INTRODUCTION

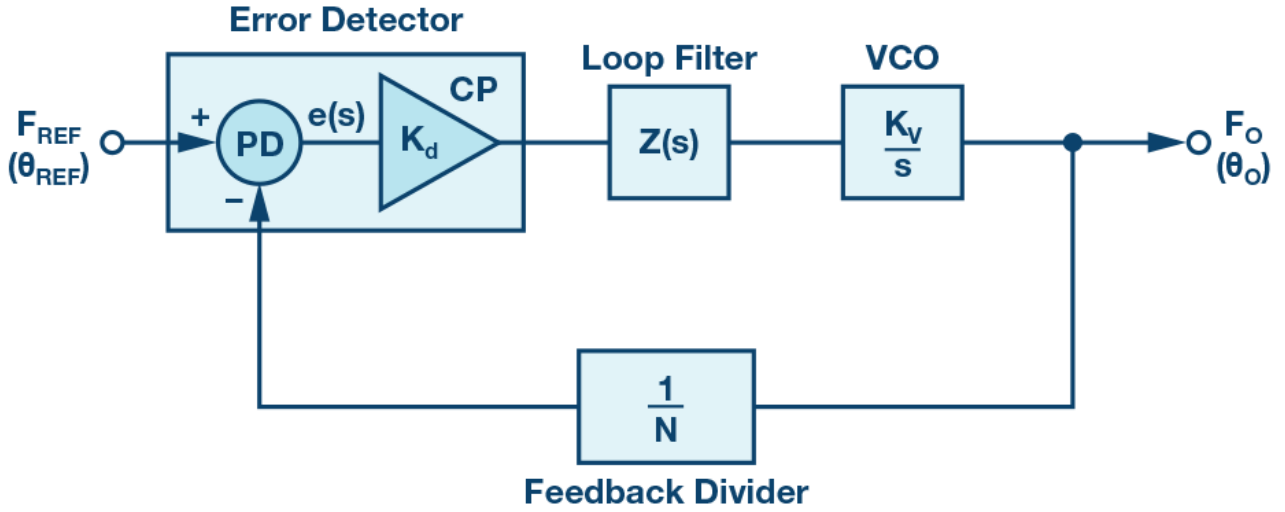


Figure 1 — PLL Block Diagram

Phase-Locked Loops (PLLs) are very important building blocks in many electronic systems, especially in modern System-on-Chip (SoC) designs. They are mainly used to generate clock signals or to change one frequency into another. This is needed for both digital circuits and communication systems, such as wireless transmitters and receivers.

As technology gets faster and more advanced, the demand for higher clock speeds and better timing performance keeps growing. To meet this demand, engineers often use Analog and Mixed-Signal PLLs (AMS-PLLs). These types of PLLs are good at generating accurate and low-noise clock signals, even at high frequencies.

This project focuses on a **Type-II PFD-CP (Phase-Frequency Detector and Charge Pump) Fractional-N PLL**, which includes key blocks like:

- \* A Phase-Frequency Detector (PFD)
- \* A Charge Pump (CP)
- \* A Loop-Pass Filter (LPF)
- \* A Voltage-Controlled Oscillator (VCO)
- \* A Frequency Divider

In this design, a **Fractional-N PLL** is created using a **mod-16 binary counter** instead of a delta-sigma modulator. This allows the PLL to divide the frequency in steps, which helps achieve more flexible output frequencies between 2 and 16 times the input reference.

To improve integration in CMOS technology, a **Ring VCO** is used instead of an LC VCO. Ring VCOs are easier to design, take up less space, and offer a wider tuning range. However, they may have higher phase noise compared to LC VCOs.

The goal of this project is to simulate the full PLL circuit using 45 nm CMOS technology. The design must meet certain specifications such as lock time, output frequency, and power consumption. Simulation results will help verify that the PLL works as expected and locks correctly to the desired output frequency.

This report explains the design steps, shows simulation results (such as time-domain and frequency-domain behavior), and also includes ideas to improve or change the architecture for use in an Integer-N PLL.

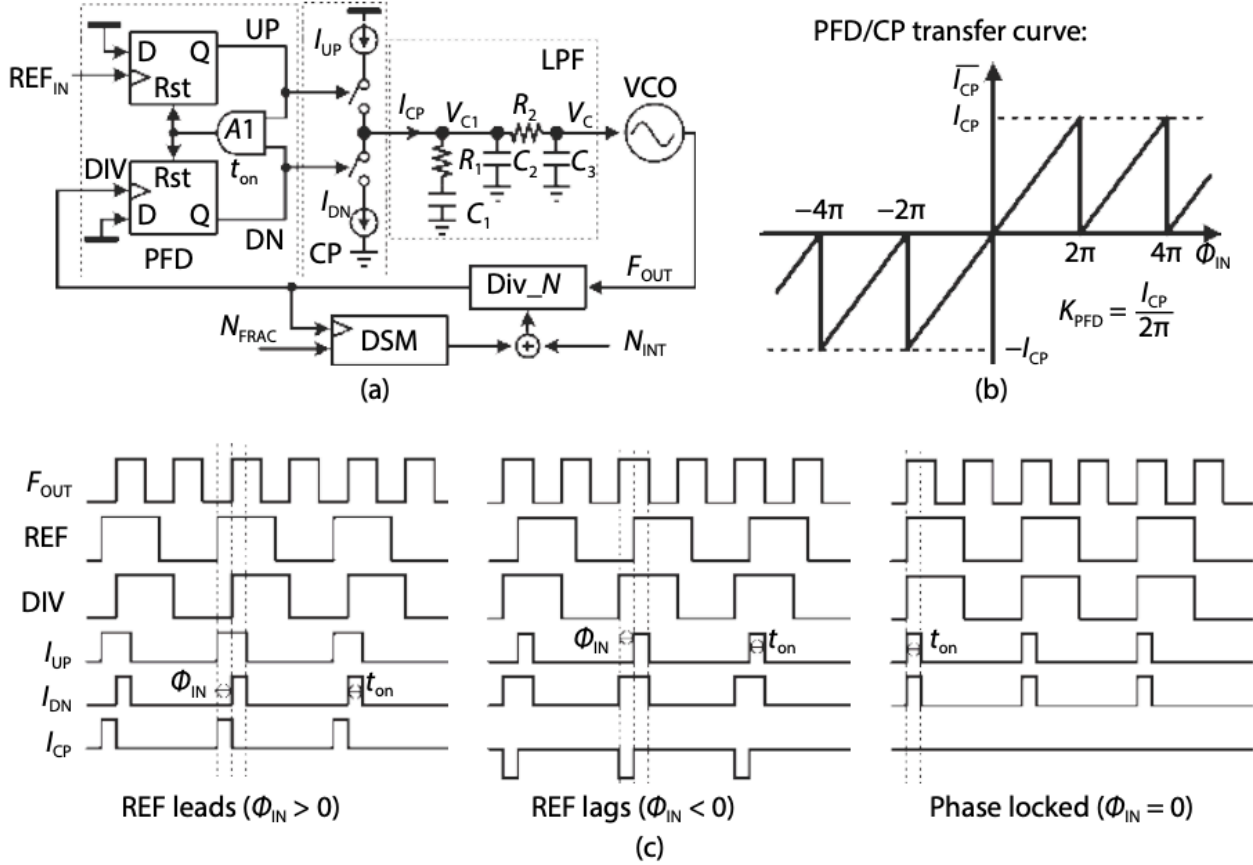


Figure 2 — (a) A basic PFD-CP (Type-II) PLL (b) Transfer curve of PFD and CP. (c) Timing diagram

## Functionality of the Type-II PFD-CP $\Delta\Sigma$ Fractional-N PLL Circuit

The proposed circuit is a Type-II Phase-Frequency Detector and Charge Pump (PFD-CP) based  $\Delta\Sigma$  Fractional-N Phase-Locked Loop (PLL) designed in 45nm CMOS technology. The architecture consists of several analog and digital sub-blocks that operate together to generate a stable high-frequency clock output from a lower frequency reference clock.

### 1. Reference Clock ( $f_{REF}$ )

This is a stable low-frequency input clock, typically generated by a crystal oscillator. In this project, the reference frequency is specified as 200 MHz. It provides the timing reference against which the PLL output is compared and locked.

### 2. Phase-Frequency Detector (PFD)

The PFD is a tri-state circuit that compares the phase and frequency of the reference clock ( $f_{REF}$ ) and the feedback clock from the frequency divider. It produces UP and DOWN signals depending on whether the reference clock leads or lags the feedback clock. These signals are used to control the charge pump.

- \* When the PLL is **unlocked** and  $\phi_{IN}$  is positive (reference leads feedback), the PFD outputs a positive pulse to the charge pump.
- \* When  $\phi_{IN}$  is negative (reference lags feedback), the PFD outputs a negative pulse.
- \* At the **locked** state,  $\phi_{IN} = 0$ , and the PFD outputs are idle.

### 3. Charge Pump (CP)

The CP translates the digital UP and DOWN signals from the PFD into analog current pulses. These pulses are then integrated by the loop-pass filter to produce a control voltage ( $V_C$ ) for the VCO.

- \* Positive UP pulses inject current to charge the loop filter, increasing  $V_C$  and hence the VCO frequency.
- \* Negative DOWN pulses sink current, reducing  $V_C$  and hence the VCO frequency.

### 4. LoOp Pass Filter (LPF)

This is typically an RC or RLC filter that integrates the CP current pulses and removes high-frequency noise. It converts the pulsed current into a stable control voltage ( $V_C$ ) which drives the VCO. The LPF also determines the loop bandwidth and damping factor of the PLL.

### 5. Voltage-Controlled Oscillator (VCO)

The VCO generates the output clock signal. Its frequency is controlled by the voltage  $V_C$  from the LPF.

- \* In this design, a **Ring VCO** is used instead of an LC VCO to reduce area and complexity.
- \* Ring VCOs offer a wider tuning range, but with higher phase noise compared to LC VCOs.

### 6. Frequency Divider with $\Delta\Sigma$ Modulator (DSM)

The output of the VCO is divided by a variable division ratio using a divider controlled by a digital  $\Delta\Sigma$  modulator.

- \* This enables fractional division, allowing finer resolution in output frequency synthesis.
- \* The DSM rapidly toggles the division ratio between two adjacent integers, averaging to a fractional value.

### 7. Overall System Functionality

The PLL operates as a negative-feedback control system:

1. The reference clock is compared with the divided feedback clock by the PFD.
2. The PFD outputs control the charge pump, which adjusts the control voltage.
3. The LPF smooths this control voltage and drives the VCO.
4. The VCO output is divided and fed back to the PFD.
5. The loop continues to adjust until the VCO output frequency  $f_{VCO}$  equals  $N \cdot f_{REF}$ , where  $N$  is the (possibly fractional) division ratio.

## 2 DESIGN METHODOLOGY AND DETAILS OF THE CALCULATION AND DEVICE SIZES OF A PFD-CP TYPE-II $\Delta\Sigma$ FRACTIONAL-N PHASE LOCKED LOOP SIMULATION

### 2.1 PLL Design Specifications

PLL Design Specifications

| Parameter                                             | Project Specification                 |
|-------------------------------------------------------|---------------------------------------|
| Technology [Min. length of transistors ( $L_{min}$ )] | 45 nm CMOS                            |
| Supply voltage $V_{DD}$                               | 1 V                                   |
| Nominal input common-mode voltage $V_{DD}/2$          | 0.5 V                                 |
| Reference frequency $f_{REF}$                         | 200 MHz                               |
| Fractional division ratio $N$                         | $2 \leq N \leq 16$                    |
| Output frequency range $f_{OUT}$                      | $f_{OUT,min}$ GHz – $f_{OUT,max}$ GHz |
| PLL lock time $t_{lock}$                              | $t_{lock,min}$ ps – $t_{lock,max}$ ps |
| Overall average DC power consumption $P_{total,avg}$  | ? $\mu$ W                             |

### 2.2 Pre-Calculations of a PFD-CP Type-II $\Delta\Sigma$ Fractional-N Phase Locked Loop Simulation

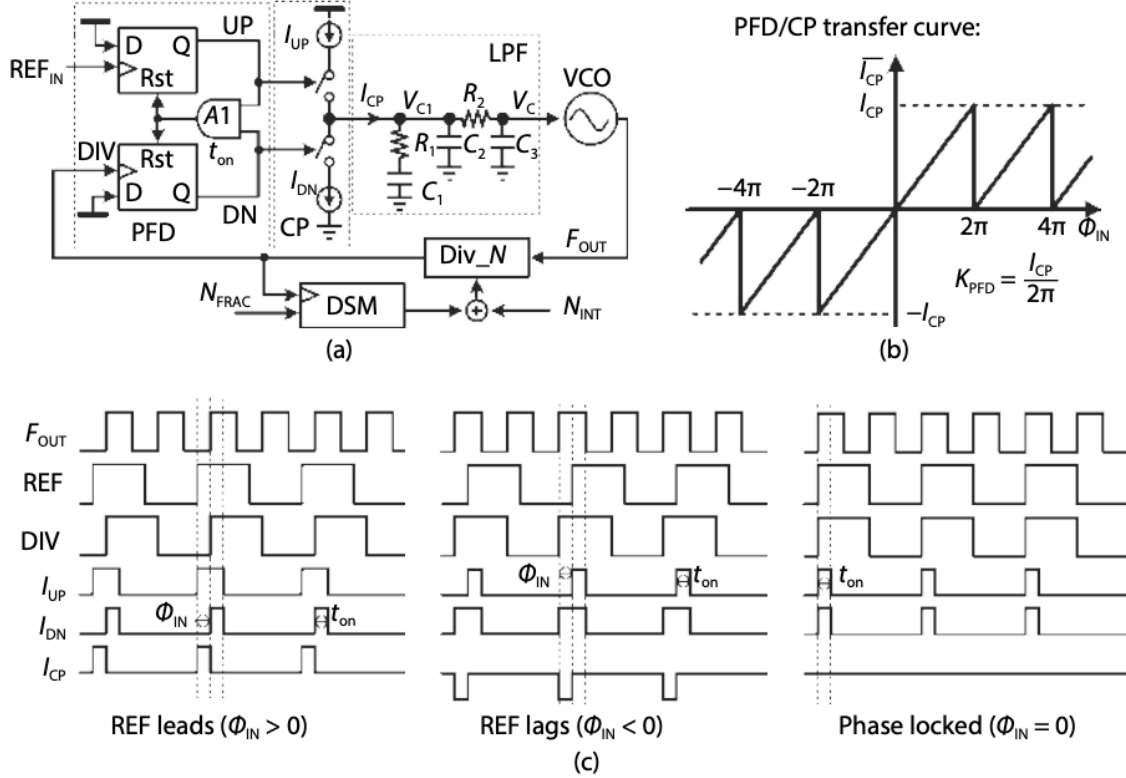


Figure 3 — A PFD-CP Type-II  $\Delta\Sigma$  Fractional-N Phase Locked Loop Simulation

## Design Methodology and Device Sizes

The design of a Type-II  $\Delta\Sigma$  Fractional-N Phase-Locked Loop (PLL) in 45 nm CMOS involves several key steps, incorporating analog and mixed-signal blocks. Here's a structured methodology tailored to this project:

### 1. Block-Level Design

#### 2.2.1 Phase Frequency Detector (PFD)

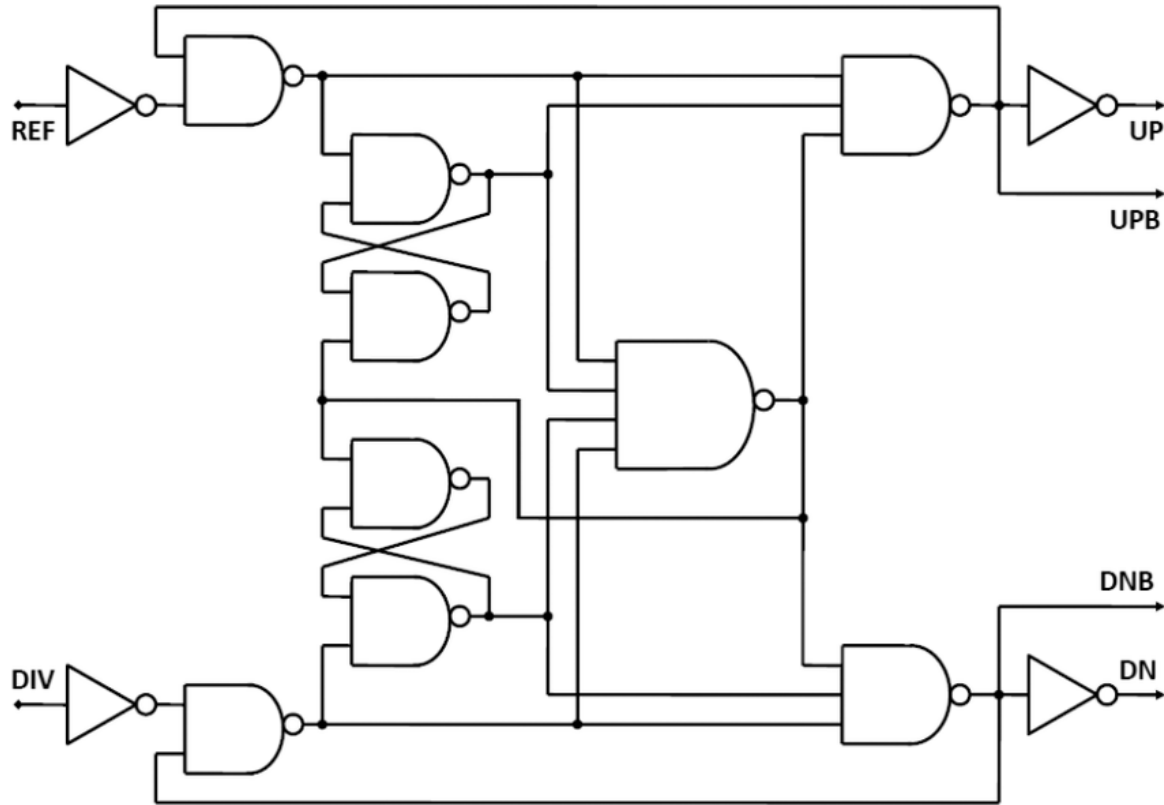


Figure 4 — PFD

Source :- [PFD Circuit](#)

- \* Tri-state PFD with reset to reduce dead zone.
- \* Key parameters: propagation delay and dead zone minimization.

### PFD Functionality Explanation

This PFD has:

#### Inputs:

- \* **REF** — the reference clock input
- \* **DIV** — the feedback clock input (from divider output)

#### Outputs:

- \* **UP, DN** — these are the charge pump control signals.

#### Working:

- \* If REF edge arrives before DIV, the PFD generates a high UP pulse.
- \* If DIV edge arrives before REF, the PFD generates a high DN pulse.
- \* If they arrive simultaneously, no output pulse is generated (ideal lock condition).

### Internal Blocks:

- \* Two D flip-flops (made using NAND and NOT gates) — one for REF, one for DIV.
- \* One Reset Logic (AND gates) to detect when both outputs (UP and DN) are high and reset both flip-flops.

Thus,

$$\text{UP pulse width} \propto \text{phase difference (REF leads DIV)}$$

$$\text{DN pulse width} \propto \text{phase difference (DIV leads REF)}$$

This tri-state behavior ensures minimal dead zone and fast locking.

### Device Sizing for 45 nm CMOS

Since the technology is 45 nm, and this circuit operates at high frequencies (up to  $\sim 3.2$  GHz), we require fast gates.

### Typical Sizing:

| Block                                          | Devices                                                                                    |
|------------------------------------------------|--------------------------------------------------------------------------------------------|
| Inverter (inside flip-flop and output drivers) | NMOS: $1\ \mu\text{m}/45\ \text{nm}$<br>PMOS: $2\ \mu\text{m}/45\ \text{nm}$               |
| NAND gate                                      | NMOS (each): $1\ \mu\text{m}/45\ \text{nm}$<br>PMOS (each): $2\ \mu\text{m}/45\ \text{nm}$ |
| AND gate (reset logic)                         | NMOS (each): $1\ \mu\text{m}/45\ \text{nm}$<br>PMOS (each): $2\ \mu\text{m}/45\ \text{nm}$ |
| Buffer stages (driving UP/DN)                  | NMOS: $1\ \mu\text{m}/45\ \text{nm}$<br>PMOS: $2\ \mu\text{m}/45\ \text{nm}$               |

All transistors use minimum length  $L = 45\ \text{nm}$ .

**Note:** PMOS are wider than NMOS because PMOS mobility is about 2–3x lower than NMOS.

Buffers on UP/DN signals are slightly upsized to drive the charge pump inputs fast and sharply.

### Important Calculations

#### Maximum Frequency

Since the target is  $f_{\text{REF}} = 200\ \text{MHz}$  and VCO up to 3.2 GHz, the PFD must work reliably up to 3.2 GHz.

Thus, the PFD maximum operating frequency  $f_{\text{max}}$ :

$$f_{\text{max}} \approx \frac{1}{2t_{\text{setup}}}$$

where  $t_{\text{setup}}$  is the setup time of flip-flop plus reset delay.

Typically:

$$t_{\text{setup}} + t_{\text{reset}} \sim 50\ \text{ps}$$

Thus:

$$f_{\text{max}} \approx 10\ \text{GHz}$$

which is comfortably safe.



## Functionality Explanation

This circuit is a high-swing NMOS-switched charge pump. It is designed to:

- \* Provide matched UP and DOWN currents.
- \* Increase output voltage swing (even close to VDD and GND).
- \* Reduce mismatch between sourcing and sinking currents (important for low jitter and low spurs).

## Key Blocks

- \* **Current Source:** A bias current source ( $I_{CP} = 80 \mu A$ ) supplies a constant current.
- \* **PMOS Current Mirrors (MP1–MP5):** Mirror and split the bias current to the upper branch.
- \* **NMOS Current Mirror (MN1–MN3):** Mirror the current for the sinking (down) branch.
- \* **Switching Transistors (MN5, MN6, MN7, MN8):** Turned ON/OFF by the PFD outputs (UPB, DN, DN, DNB).
- \* **High-Swing Cascode Structure:** MP2 and MP3 (for sourcing) and MN2 and MN3 (for sinking) form cascoded current sources to increase output resistance, improving matching.
- \* **Output Node ( $V_{con}$ ):** Net result of charging or discharging depending on PFD control.

## Transistor Device Sizes

| Transistor | $W/L$ ( $\mu m/\mu m$ ) |
|------------|-------------------------|
| MP1        | 4 / 0.5                 |
| MP2        | 11 / 0.2                |
| MP3        | 11 / 0.2                |
| MP4        | 11 / 0.2                |
| MP5        | 11 / 0.2                |
| MP6        | 20 / 0.12               |
| MP7        | 20 / 0.12               |
| MP8        | 0.5 / 0.2               |
| MP9        | 0.5 / 0.2               |
| MN1        | 2 / 0.2                 |
| MN2        | 2 / 0.2                 |
| MN3        | 2 / 0.2                 |
| MN4        | 2 / 0.12                |
| MN5        | 2 / 0.12                |
| MN6        | 2 / 0.12                |
| MN7        | 2 / 0.12                |
| MN8        | 10 / 0.2                |
| MN9        | 10 / 0.2                |
| MN10       | 10 / 0.2                |
| MN11       | 10 / 0.2                |

## Notes:

- \* PMOS devices are generally wider to compensate for lower hole mobility.
- \* Cascode devices (MP2, MP3, MN2, MN3) are wider to ensure minimal voltage drop and maintain high output impedance.
- \* Switches MN7, MN8 are very wide ( $10 \mu m$ ) to minimize voltage drop and resistive loss when turned ON.

## Important Calculations

### Matching Error

Mismatch causes phase noise and spurs. The mismatch  $\delta$  between sourcing and sinking currents should be:

$$\delta = \frac{|I_{up} - I_{down}|}{I_{cp}} \leq 1\%$$

where  $I_{cp} = 80 \mu A$ .

Thus:

$$|I_{up} - I_{down}| \leq 0.8 \mu A$$

### Output Voltage Range

Thanks to the high-swing cascodes, the output ( $V_{con}$ ) swing is improved:

$$V_{out,max} \approx V_{DD} - V_{SG} - V_{DS,sat}$$

$$V_{out,min} \approx V_{DS,sat}$$

Given:

$$* V_{SG} \approx 0.25 V$$

$$* V_{DS,sat} \approx 0.1 V$$

Thus:

$$V_{out,max} \approx 1 V - 0.25 V - 0.1 V = 0.65 V$$

$$V_{out,min} \approx 0.1 V$$

This is sufficient because the VCO control voltage typically varies between  $0.1 V$  and  $0.9 V$ .

### Charge Pump Gain

The charge pump gain  $K_{CP}$  is:

$$K_{CP} = \frac{I_{cp}}{2\pi}$$

Given  $I_{cp} = 80 \mu A$ :

$$K_{CP} = \frac{80 \times 10^{-6}}{2\pi} \approx 12.7 \mu A/rad$$

This value is used in the PLL open-loop gain calculation.

### Final values for charge pump

| Item                 | Value                                                                             |
|----------------------|-----------------------------------------------------------------------------------|
| Technology           | 45 nm CMOS                                                                        |
| $V_{DD}$             | 1 V                                                                               |
| $I_{CP}$             | 80 $\mu A$                                                                        |
| PMOS Sizes           | MP1: 4/0.5; MP2, MP3, MP4, MP5: 11/0.2;<br>MP6, MP7: 20/0.12; MP8, MP9: 0.5/0.2   |
| NMOS Sizes           | MN1, MN2, MN3: 2/0.2; MN4, MN5, MN6, MN7: 2/0.12;<br>MN8, MN9, MN10, MN11: 10/0.2 |
| Matching Error       | < 1%                                                                              |
| Output Voltage Range | $\sim 0.1 V$ to $0.65 V$                                                          |
| $K_{CP}$             | $\sim 12.7 \mu A/rad$                                                             |

### 2.2.3 Loop Filter (LPF)

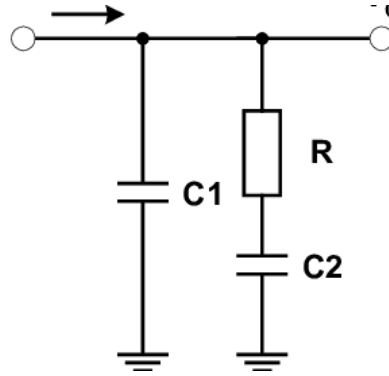


Figure 6 — LPF

source : [LPF circuit](#)

- \* Design a second-order passive filter.
- \* Select component values to balance loop bandwidth and stability:

$$R_1 = 1\text{ k}\Omega, \quad C_1 = 5\text{ pF}, \quad C_2 = 1\text{ pF}$$

### 2.2.4 Ring VCO

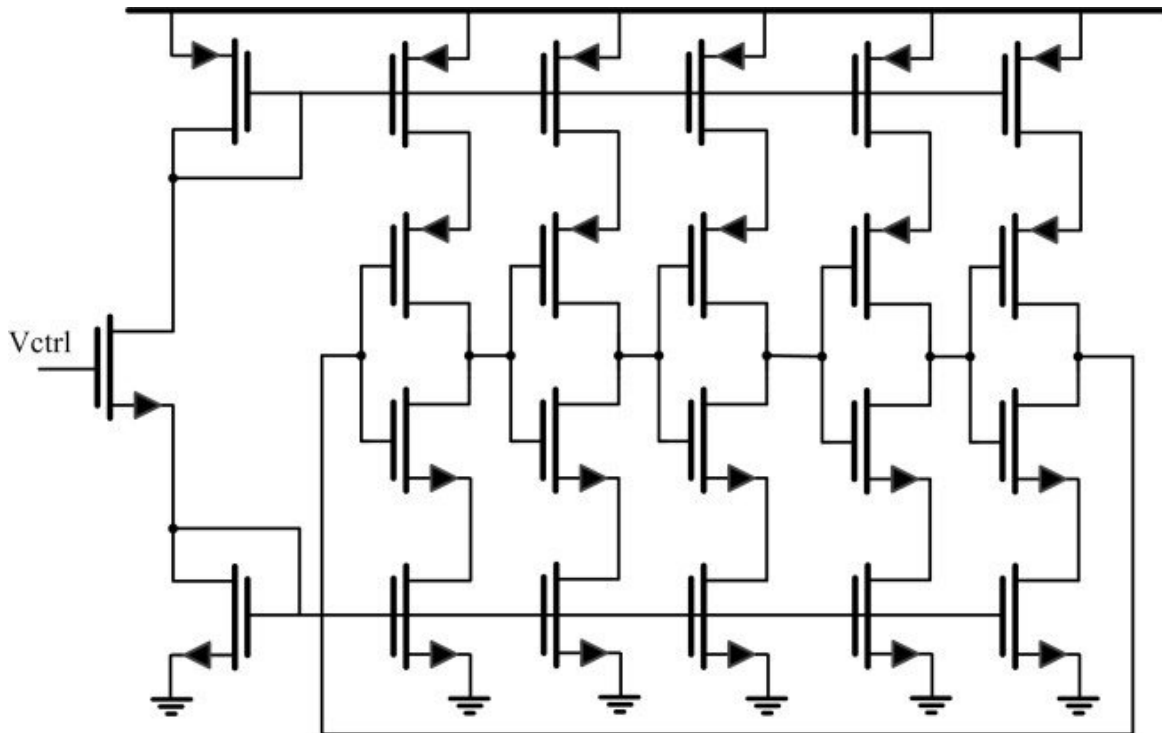


Figure 7 — RING VCO

source : [VCO circuit](#)

#### Functionality of the Ring VCO

- \* Implement a 5-stage differential ring oscillator.
- \* Frequency tuning range should cover 0.4 GHz to 3.2 GHz (based on  $N = 2$  to 16 with  $f_{\text{REF}} = 200\text{ MHz}$ ).

The Ring VCO consists of an odd number (here, 5) of inverter stages connected in a ring (feedback from last output to the first input).

- \* Each inverter introduces a delay, and the total delay around the ring determines the oscillation frequency.
- \* The delay — and thus the frequency — is controlled using a control voltage ( $V_{ctrl}$ ).
- \* NMOS and PMOS transistors are used as current sources controlled by  $V_{ctrl}$  (common in current-starved inverters).

### Basic Working Principle

- \* Inverter delay is proportional to the supply current.
- \*  $V_{ctrl}$  changes the current through each inverter stage.
- \* Higher  $V_{ctrl} \rightarrow$  higher current  $\rightarrow$  faster switching  $\rightarrow$  higher frequency.
- \* Lower  $V_{ctrl} \rightarrow$  lower current  $\rightarrow$  slower switching  $\rightarrow$  lower frequency.

The oscillation frequency  $f_{osc}$  is approximately:

$$f_{osc} = \frac{1}{2Nt_d}$$

where:

- \*  $N$  = number of stages (here, 5),
- \*  $t_d$  = delay per inverter stage.

### Device Sizing (General Guide)

To design properly, a balance must be struck between:

- \* Speed (lower delay),
- \* Power consumption (lower current),
- \* Swing (rail-to-rail output if needed).

Typical initial sizing approach:

- \* Use an NMOS and PMOS pair per inverter.
- \* Assume minimum channel length for the technology (e.g.,  $L = 45$  nm).

Widths:

- \* NMOS width  $W_n$ :  $2\text{--}5 \times$  minimum width,
- \* PMOS width  $W_p$ :  $1.5 \times W_n$  to  $2 \times W_n$  (to compensate for lower hole mobility).

sizing (for 45nm technology):

$$L = 45 \text{ nm}$$

$$W_n = 1 \text{ }\mu\text{m}, \quad W_p = 2 \text{ }\mu\text{m}$$

Later adjustment is needed based on simulation results (targeting desired  $f_{osc}$ , power, tuning range, etc.).

## Other Calculations

Suppose the target is:

$$f_{osc} = 1 \text{ GHz}$$

Then the delay per stage needed:

$$t_d = \frac{1}{2Nf_{osc}} = \frac{1}{2 \times 5 \times 1 \times 10^9} = 100 \text{ ps}$$

Thus, each inverter must have a delay of approximately 100 ps.

Controlling:

- \* Drive current  $I_d$  of NMOS/PMOS,
- \* Load capacitance  $C_L$  per stage.

Approximate relation:

$$t_d \approx \frac{C_L V_{DD}}{I_d}$$

where:

- \*  $C_L$  includes parasitic capacitance (input capacitance of next stage + wiring),
- \*  $I_d$  depends on sizing and overdrive voltage.

Adjust  $W/L$  to control current.

## Extra Design Notes

- \* The number of stages must be odd (for oscillation).
- \*  $V_{ctrl}$  often biases a current source that sets inverter strength.
- \* Add an output buffer if the VCO output needs to drive large loads.

Simulation: After schematic, simulate VCO's  $f_{osc}$  vs  $V_{ctrl}$  curve (known as the VCO Gain,  $K_{VCO}$ ):

$$K_{VCO} = \frac{df_{osc}}{dV_{ctrl}} \quad [\text{Hz/V}]$$

sectionFinal Parameters for VCO

| Parameter                  | Suggested Initial Value |
|----------------------------|-------------------------|
| Technology                 | 45nm CMOS               |
| NMOS Width $W_n$           | 1 $\mu\text{m}$         |
| PMOS Width $W_p$           | 2 $\mu\text{m}$         |
| Channel Length $L$         | 45 nm                   |
| Supply Voltage $V_{DD}$    | 1V                      |
| Target Frequency $f_{osc}$ | 400 MHz – 3.2 GHz       |
| Number of Stages $N$       | 5                       |

### 2.2.5 Frequency Divider

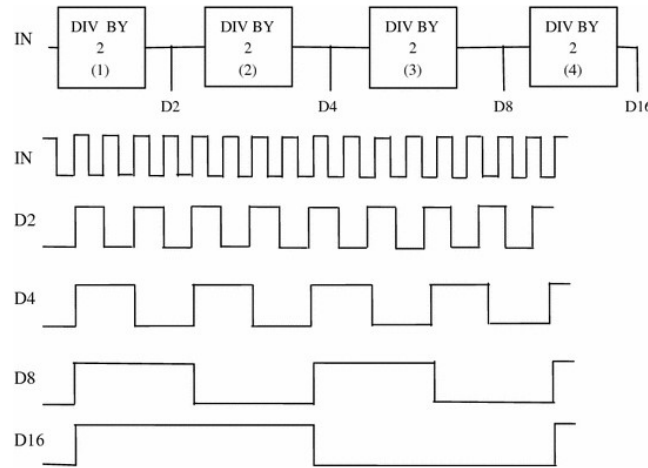


Figure 8 — Frequency Divider by 2,4,8,16

- \* Synchronous 4-bit binary counter using D flip-flops.
- \* Designed to handle VCO frequencies up to 3.2 GHz.

To design and implement a frequency divider circuit capable of dividing an input clock by programmable factors: **2, 3, 4, 5, 6, 8, 10, 12, 15, and 16**, using **D flip-flops** and counter logic suitable for use in a Phase-Locked Loop (PLL) system.

#### Explanation of functionality of Frequency Divider

##### Binary Divider Design ( $\div 2$ , $\div 4$ , $\div 8$ , $\div 16$ )

##### Topology

A chain of D flip-flops can implement a binary frequency divider:

- \* Each flip-flop toggles on the rising edge of the clock.
- \* The Q output of each stage serves as the clock input to the next stage.

##### Frequency Table (Assuming Input Frequency = $F$ )

| Output | Frequency   |
|--------|-------------|
| Q1     | $F \div 2$  |
| Q2     | $F \div 4$  |
| Q3     | $F \div 8$  |
| Q4     | $F \div 16$ |

##### Working Principle

- \* Counter increments on each clock cycle.
- \* When count =  $N - 1$ , the comparator triggers:
  - Counter resets to 0
  - Output flip-flop toggles (producing output clock)
- \* Effective frequency:  $F_{\text{out}} = \frac{F_{\text{in}}}{2N}$

**Divider Configuration Table (Assume  $F_{in} = 16$  MHz)**

| Divide Factor ( $N$ ) | Max Count | Toggle Interval | Output Frequency |
|-----------------------|-----------|-----------------|------------------|
| 2                     | 1         | Every 2 cycles  | 8.00 MHz         |
| 3                     | 2         | Every 3 cycles  | 5.33 MHz         |
| 4                     | 3         | Every 4 cycles  | 4.00 MHz         |
| 5                     | 4         | Every 5 cycles  | 3.20 MHz         |
| 6                     | 5         | Every 6 cycles  | 2.67 MHz         |
| 8                     | 7         | Every 8 cycles  | 2.00 MHz         |
| 10                    | 9         | Every 10 cycles | 1.60 MHz         |
| 12                    | 11        | Every 12 cycles | 1.33 MHz         |
| 15                    | 14        | Every 15 cycles | 1.07 MHz         |
| 16                    | 15        | Every 16 cycles | 1.00 MHz         |

### 2.2.6 Loop Parameter Calculation

Use the classical linear PLL model for the open-loop transfer function:

$$H(s) = \frac{K_{PD} \cdot I_{CP} \cdot Z(s) \cdot K_{VCO}}{s}$$

Where:

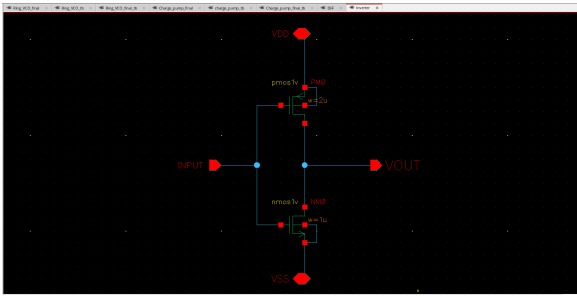
$$K_{PD} = \frac{I_{CP}}{2\pi}, \quad Z(s): \text{ Transfer function of the loop filter, } \quad K_{VCO}: \text{ VCO gain in rad/V} \cdot \text{s}$$

### 3 CIRCUIT DESIGN FOR A PFD-CP TYPE-II $\Delta\Sigma$ FRACTIONAL-N PHASE LOCKED LOOP SIMULATION

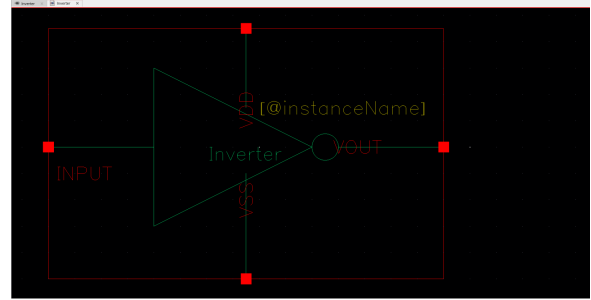
#### 3.1 Basic components

We need some basic building blocks like 2-input NAND gate, 3-input NAND gate, NOT gate/CMOS Inverter, D flip-flop with reset, and D flip-flop without reset to design a PFD, a charge pump, a VCO, and multiple frequency dividers. So, I designed them first.

##### 3.1.1 CMOS Inverter



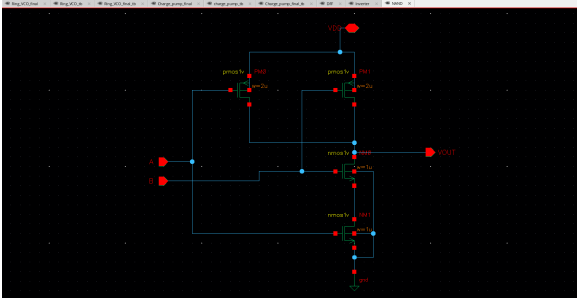
(a) Circuit Diagram



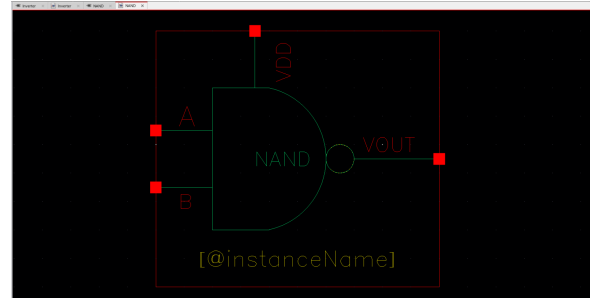
(b) Symbol

Figure 9 — CMOS Inverter

##### 3.1.2 Two-input NAND Gate



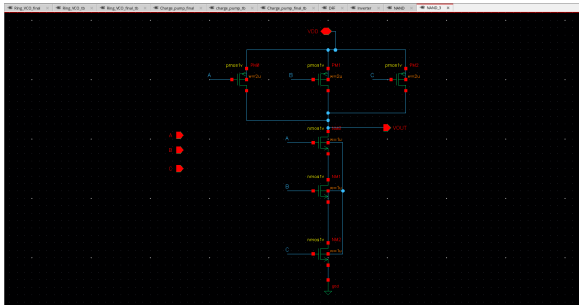
(a) Circuit Diagram



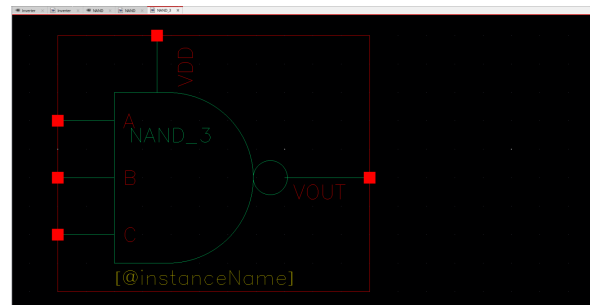
(b) Symbol

Figure 10 — Two-input NAND Gate

### 3.1.3 Three-input NAND Gate



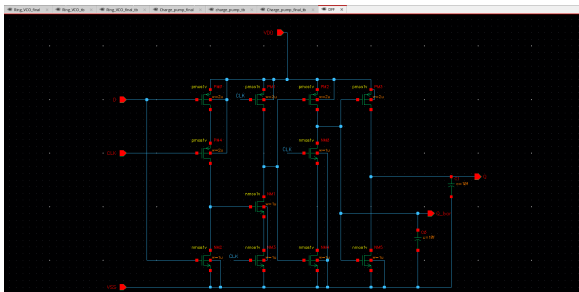
(a) Circuit Diagram



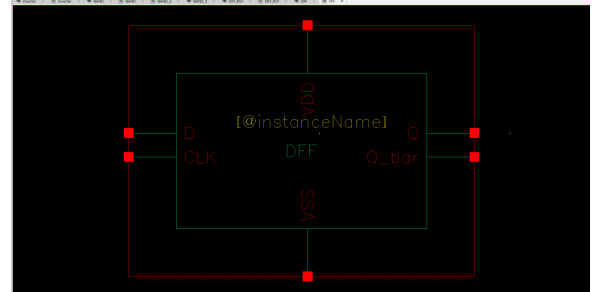
(b) Symbol

Figure 11 — Three-input NAND Gate

### 3.1.4 D Flip Flop without Resr



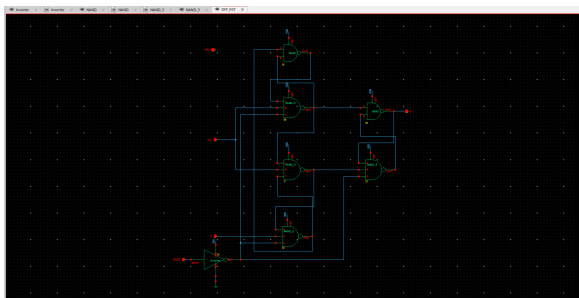
(a) Circuit Diagram



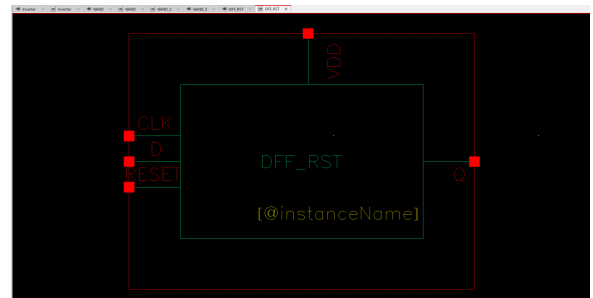
(b) Symbol

Figure 12 — D Flip Flop without Reset

### 3.1.5 D Flip Flop with Reset



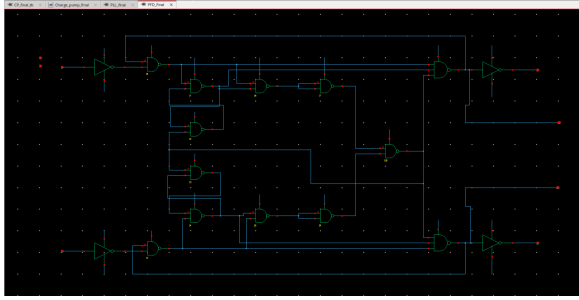
(a) Circuit Diagram



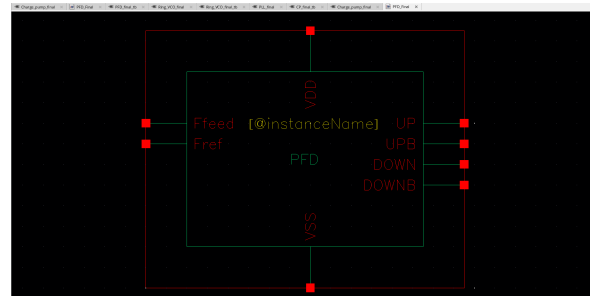
(b) Symbol

Figure 13 — D Flip Flop with Reset

### 3.2 A Phase-Frequency Detector (PFD)



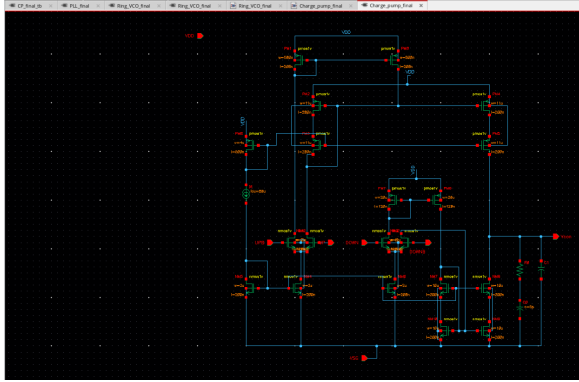
(a) Circuit Diagram



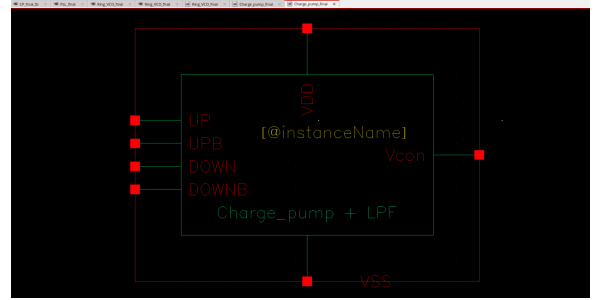
(b) Symbol

Figure 14 — A Phase-Frequency Detector (PFD)

### 3.3 A Charge Pump (CP) + a loop-Pass Filter (LPF)



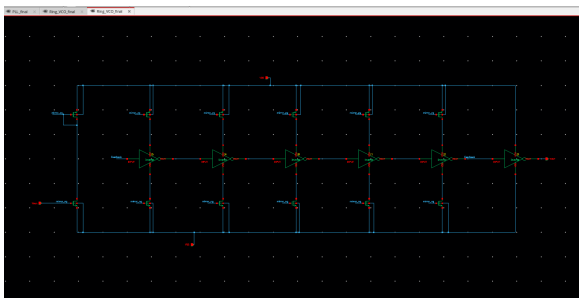
(a) Circuit Diagram



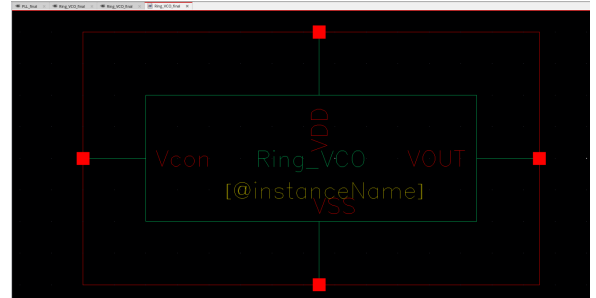
(b) Symbol

Figure 15 — A Charge Pump (CP) + A Loop-Pass Filter (LPF)

### 3.4 A Voltage-Controlled Oscillator (VCO)



(a) Circuit Diagram



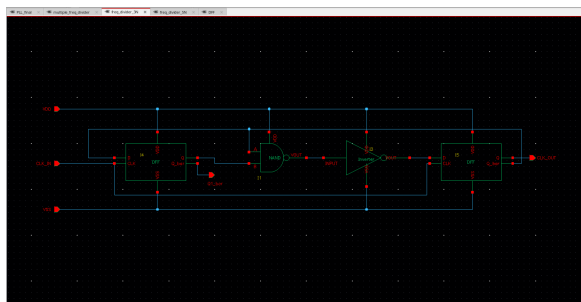
(b) Symbol

Figure 16 — A Voltage-Controlled Oscillator (VCO)

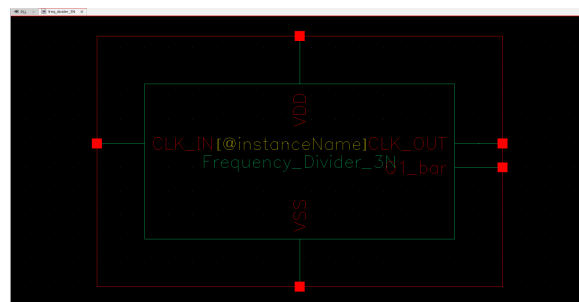
### 3.5 A Frequency Divider

The design incorporates sub-frequency dividers—specifically divide-by-3 and divide-by-5 circuits—which are cascaded to construct the overall multi-modulus frequency divider. The schematics of these sub-dividers are presented below.

### 3.5.1 Frequency divider by 3



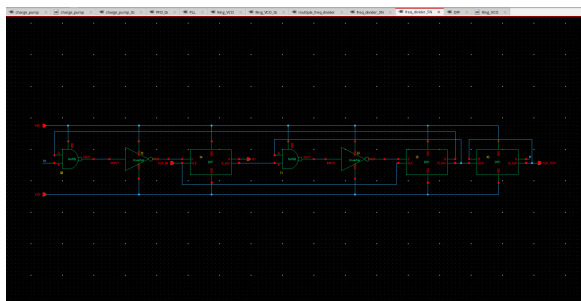
(a) Circuit Diagram



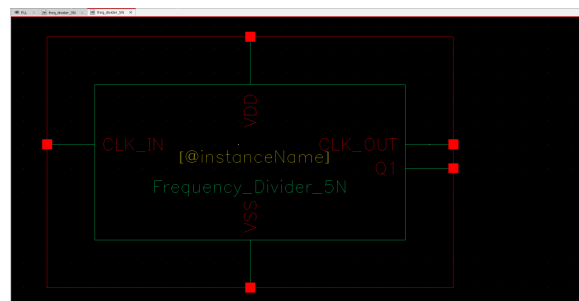
(b) Symbol

Figure 17 — A Frequency Divider by 3

### 3.5.2 Frequency divider by 5



(a) Circuit Diagram

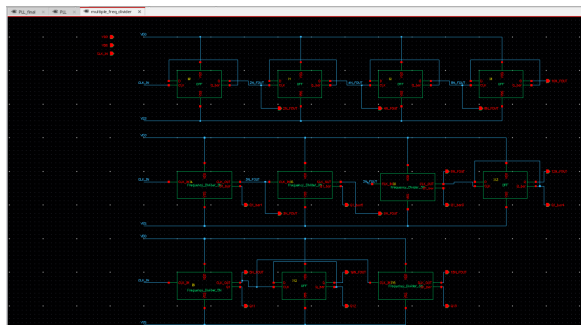


(b) Symbol

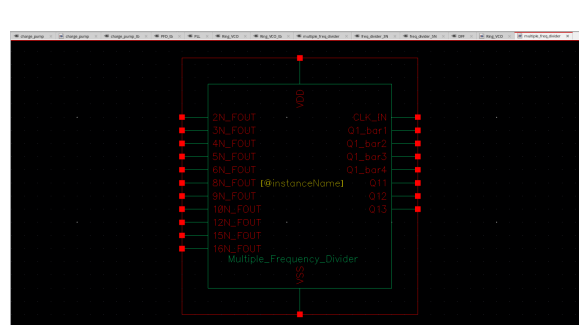
Figure 18 — A Frequency Divider y 5

### 3.5.3 Multi frequency divider

The multiple frequency divider enables division of the VCO output by factors such as 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, and 16, making it a key component in integer-N PLLs. It uses cascaded sub-divider blocks—for example, a divide-by-3 followed by a divide-by-5 to achieve divide-by-15—and is implemented using flip-flop stages with CLK\_IN and CLK\_OUT as input and output signals.



(a) Circuit Diagram



(b) Symbol

Figure 19 — A multi-frequency Divider

## 4 SIMULATION RESULTS FOR A PFD-CP TYPE-II $\Delta\Sigma$ FRACTIONAL-N PHASE LOCKED LOOP SIMULATION

### 4.1 Simulation results of a Phase-Frequency Detector (PFD)

#### 4.1.1 Test bench

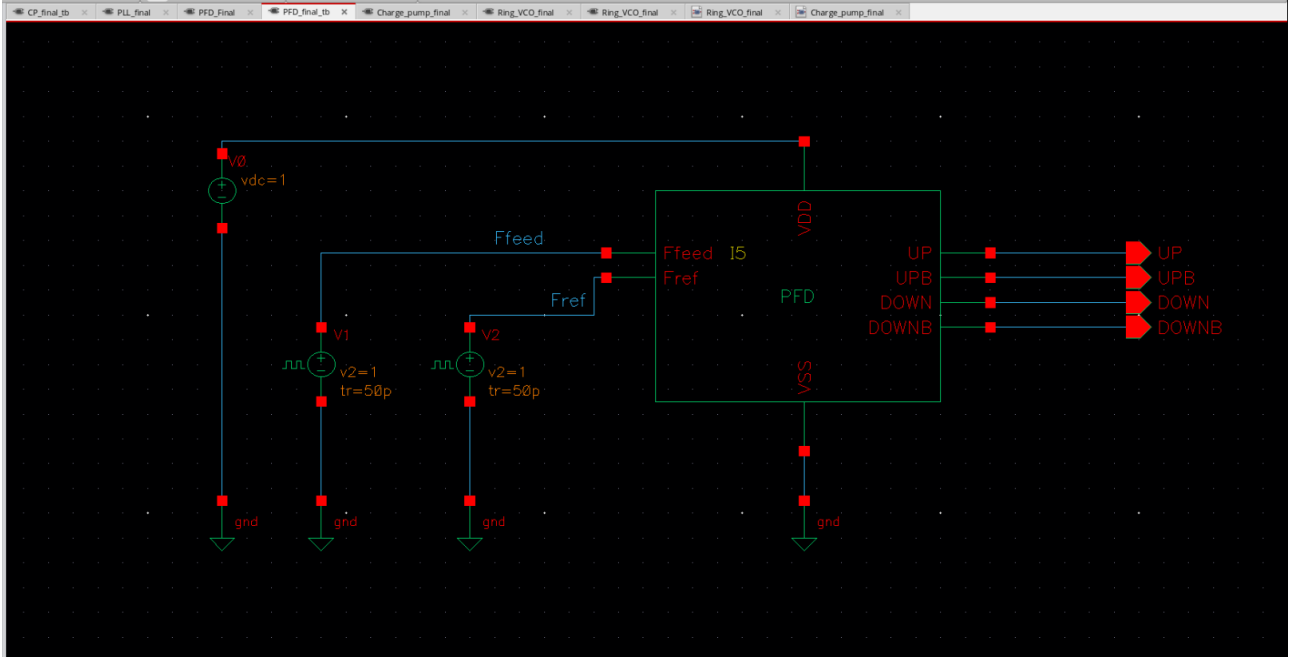


Figure 20 — Test bench for Phase-Frequency Detector (PFD)

#### 4.1.2 Transient simulation

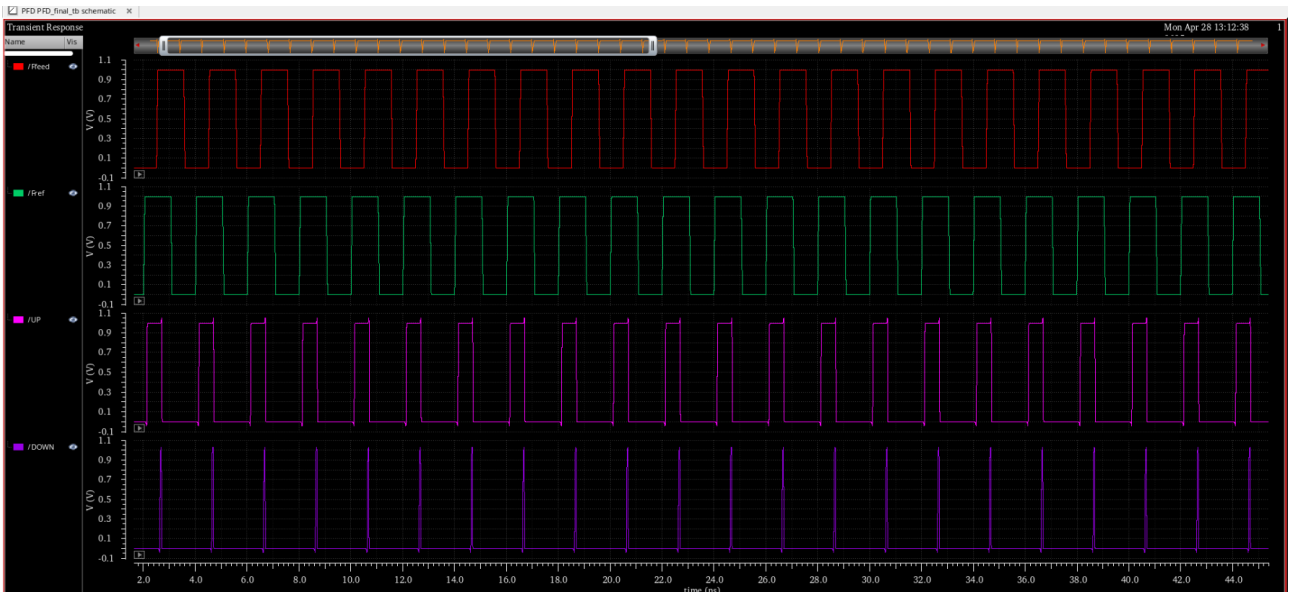


Figure 21 — Transient simulation for Phase-Frequency Detector (PFD)

## 4.2 Simulation results of a PFD + a Charge Pump (CP) + a Loop-Pass Filter (LPF)

### 4.2.1 Test bench

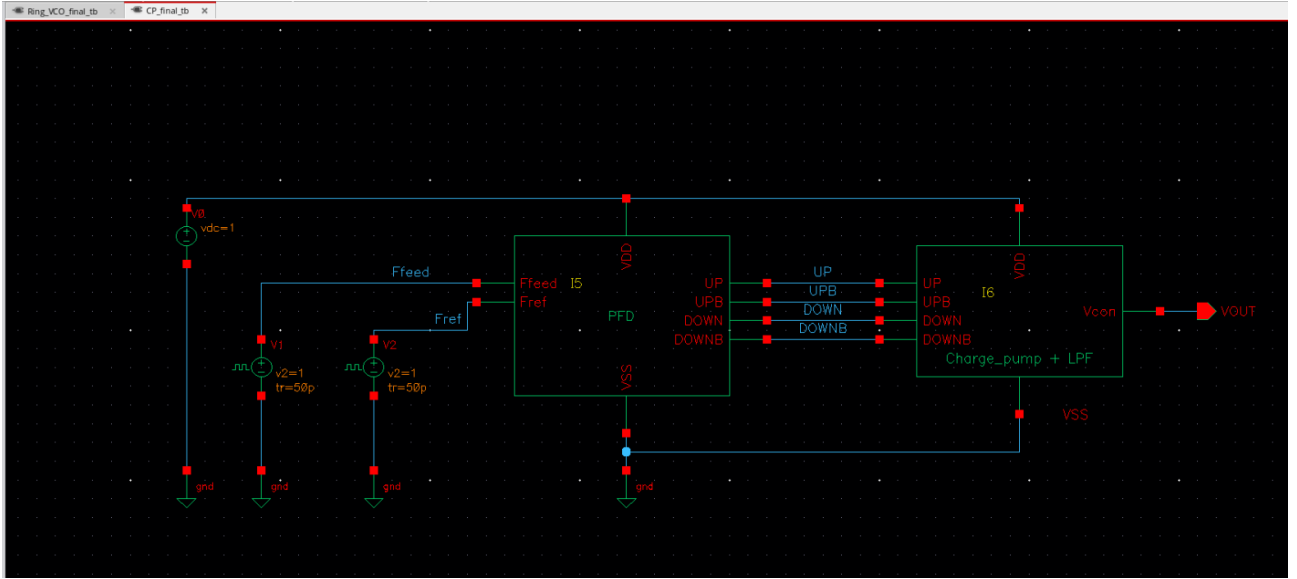


Figure 22 — Test bench

### 4.2.2 Transient simulation when $F_{ref}$ Leads $F_{feed}$

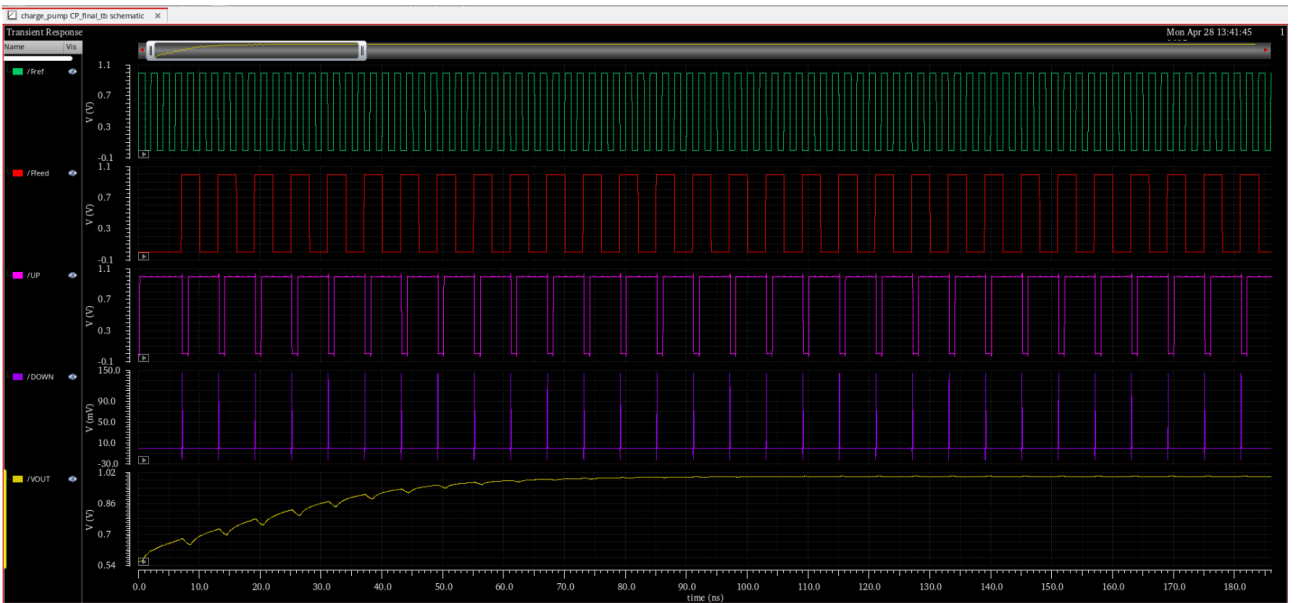


Figure 23 — Transient simulation for a PFD + a Charge Pump (CP) + a Loop-Pass Filter (LPF) when  $F_{ref}$  Leads  $F_{feed}$

### 4.2.3 Transient simulation when $F_{ref}$ Lags $F_{feed}$

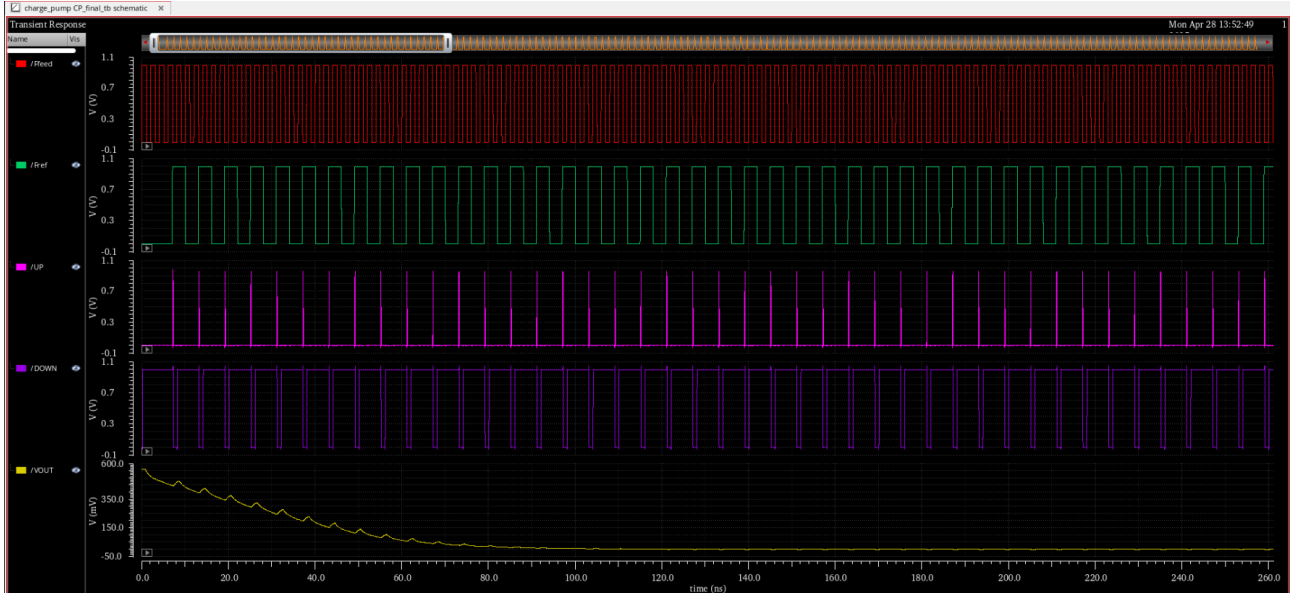


Figure 24 — Transient simulation for a PFD + a Charge Pump (CP) + a Loop-Pass Filter (LPF) when  $F_{ref}$  Lags  $F_{feed}$

## 4.3 Simulation results of a Voltage-Controlled Oscillator (VCO)

### 4.3.1 Test bench

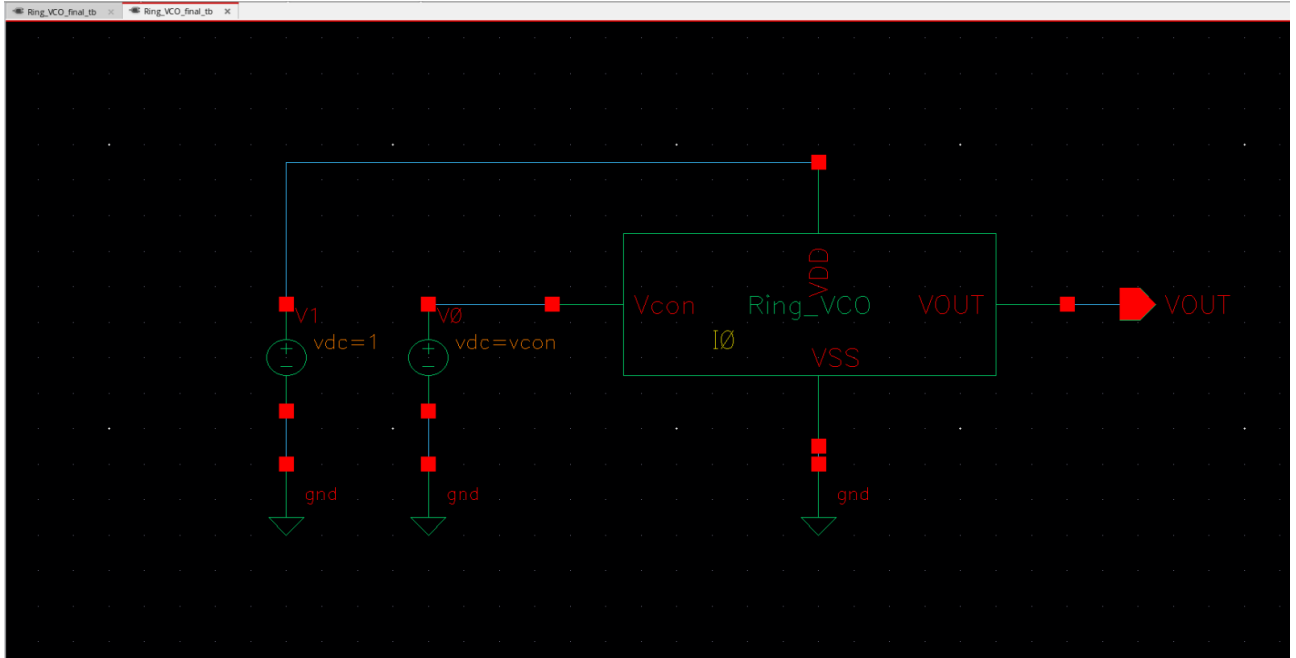


Figure 25 — Test bench for Voltage-Controlled Oscillator (VCO)

### 4.3.2 Transient simulation

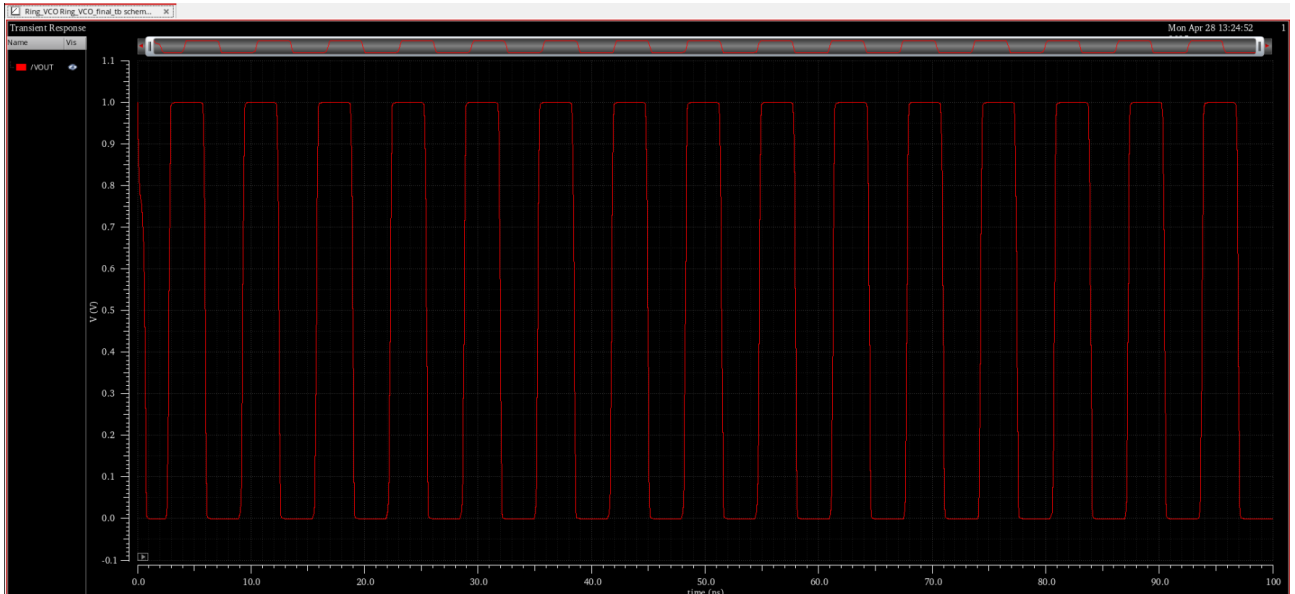


Figure 26 — Transient simulation for Voltage-Controlled Oscillator (VCO)

### 4.3.3 Parametric simulation - VCO range

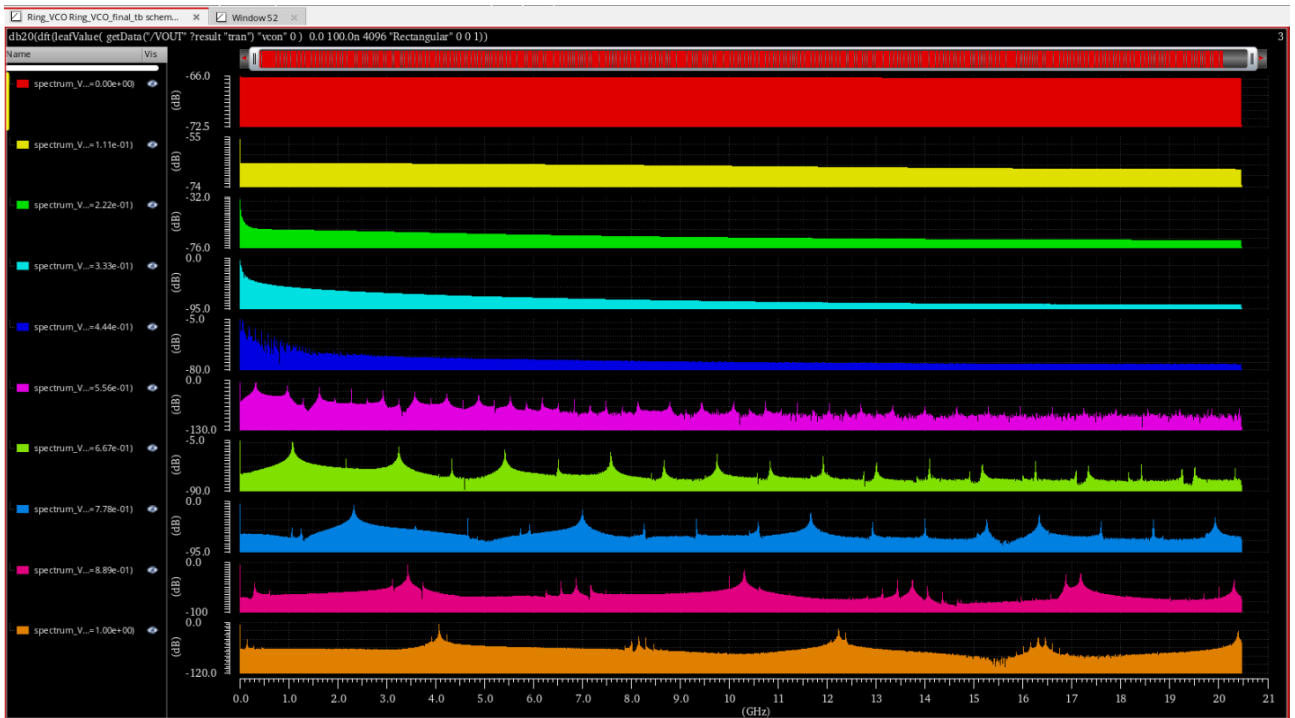


Figure 27 — Parametric simulation - FFT Results for Vctrl range from 0V to 1V for Voltage-Controlled Oscillator (VCO)

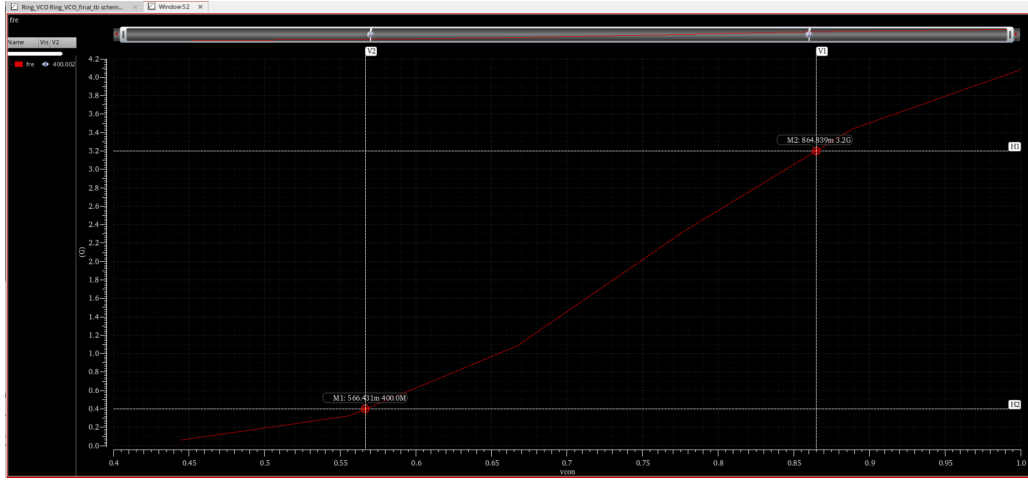


Figure 28 — Parametric simulation - VCO range for Voltage-Controlled Oscillator (VCO)

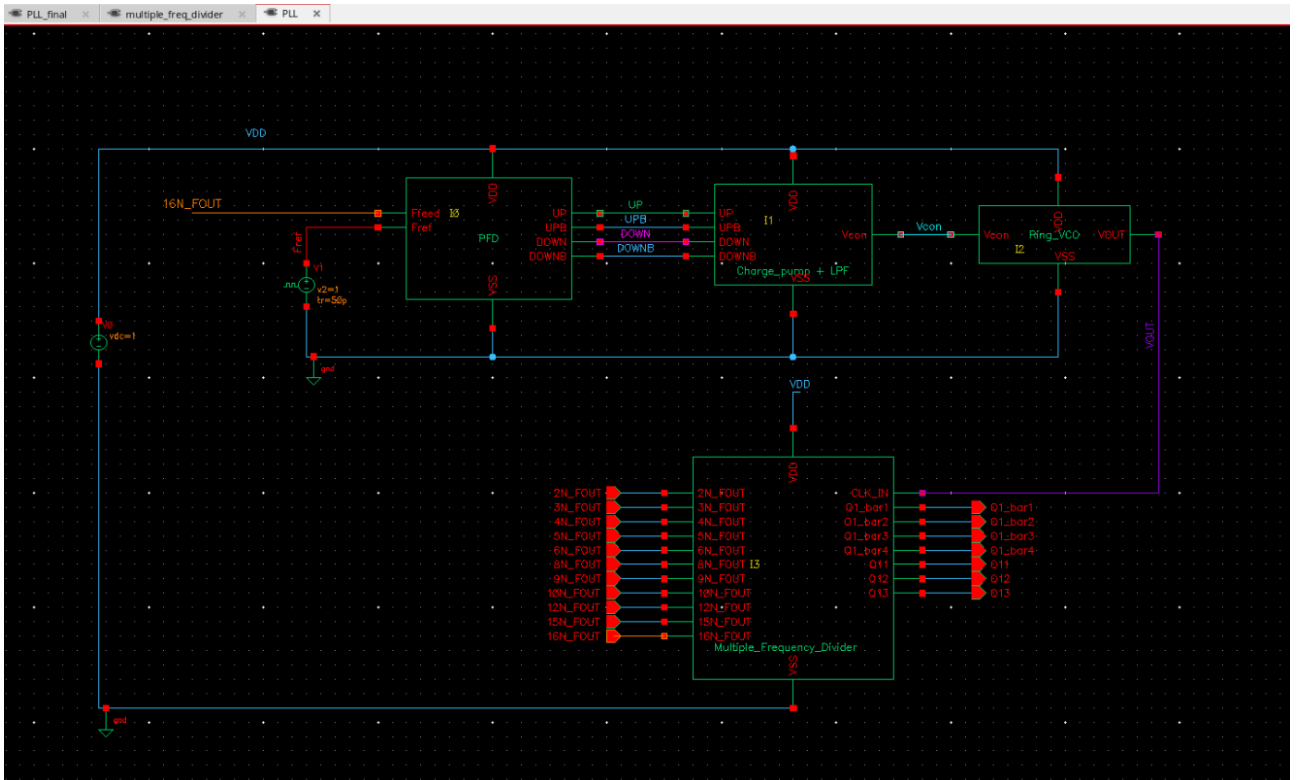
The PLL is designed to operate for division ratios  $N = 2$  to  $N = 16$ , with a reference frequency of  $f_{\text{ref}} = 200$  MHz. This results in the output frequency

$$f_{\text{vco}} = N \cdot f_{\text{ref}},$$

yielding a frequency range from 400 MHz to 3.2 GHz. According to the  $V_{\text{ctrl}}$  versus  $f_{\text{out}}$  plot of the three-stage ring VCO, a wide tuning range can be achieved by varying  $V_{\text{ctrl}}$  from 0 to 1. Based on simulation results, to achieve the full frequency range of 400 MHz to 3.2 GHz,  $V_{\text{ctrl}}$  must be adjusted between 566.43 mV and 864.839 mV.

#### 4.4 Simulation results of a PFD-CP Type-II $\Delta\Sigma$ Fractional-N Phase Locked Loop

##### 4.4.1 Test bench


 Figure 29 — Test bench of PFD-CP Type-II  $\Delta\Sigma$  Fractional-N Phase Locked Loop

#### 4.4.2 PLL Performance with frequency Divide-by-2 Configuration

##### Transient simulation

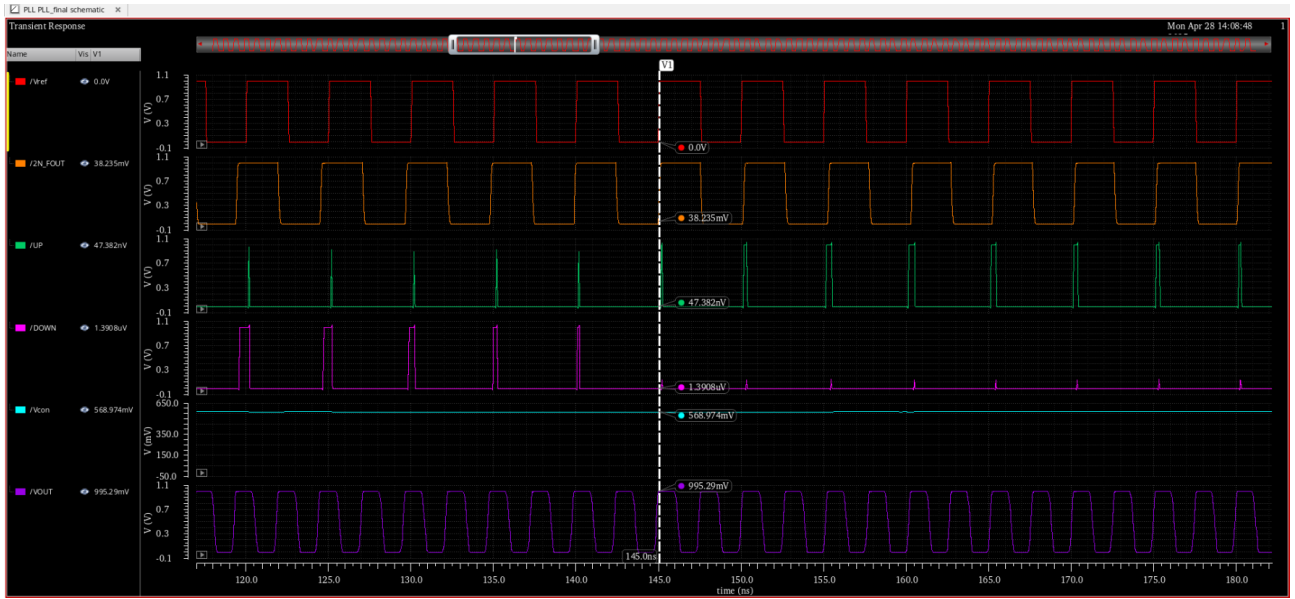


Figure 30 — Transient simulation of a PLL Performance with frequency Divide-by-2 Configuration

##### FFT spectrum

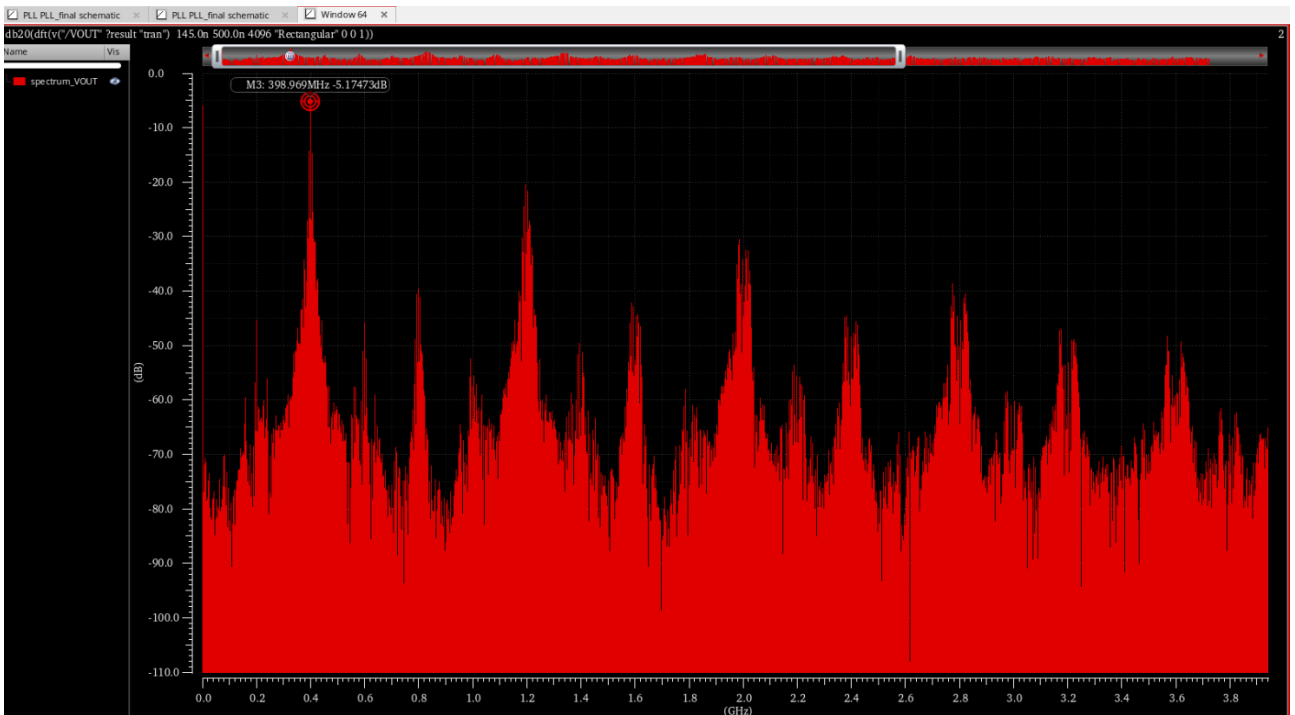


Figure 31 — FFT spectrum of a PLL Performance with frequency Divide-by-2 Configuration

### 4.4.3 PLL Performance with frequency Divide-by-3 Configuration

#### Transient simulation

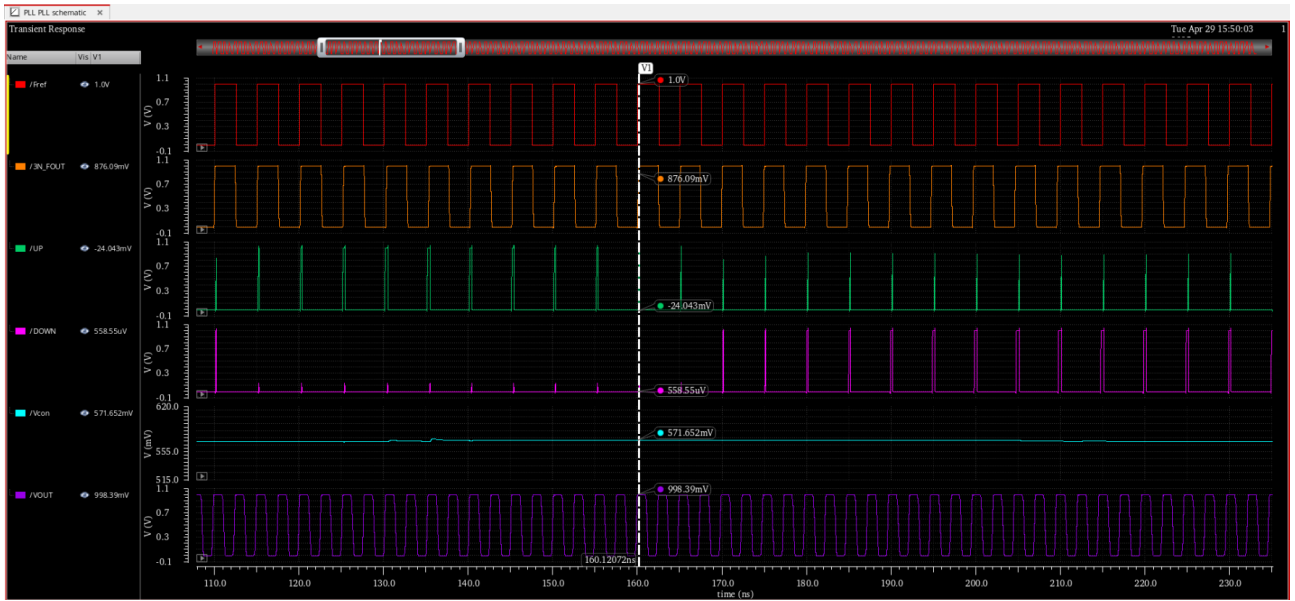


Figure 32 — Transient simulation of a PLL Performance with frequency Divide-by-3 Configuration

#### FFT spectrum

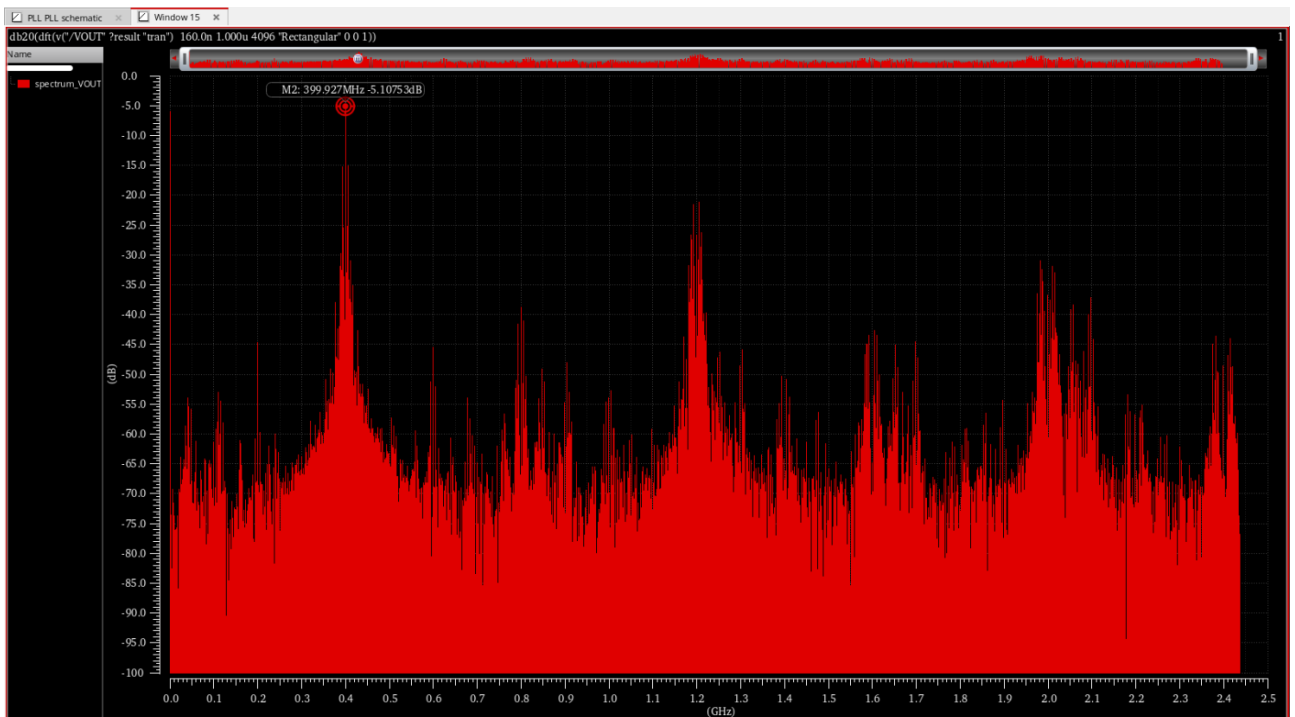


Figure 33 — FFT spectrum of a PLL Performance with frequency Divide-by-3 Configuration

#### 4.4.4 PLL Performance with frequency Divide-by-4 Configuration

##### Transient simulation

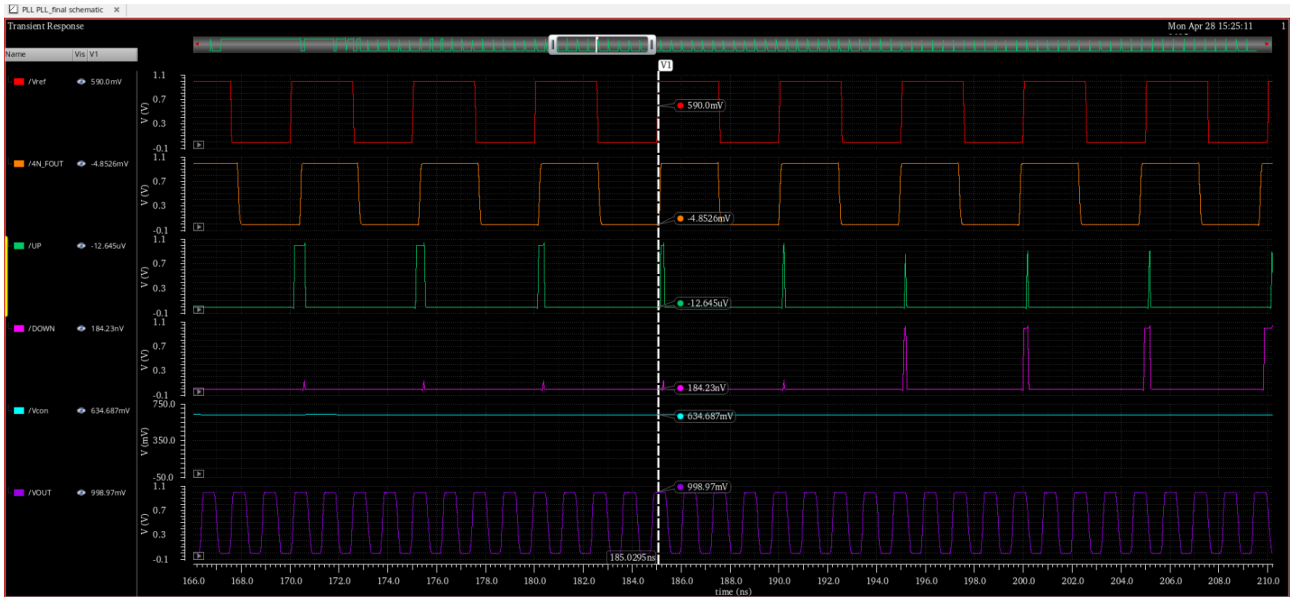


Figure 34 — Transient simulation of a PLL Performance with frequency Divide-by-4 Configuration

##### FFT spectrum

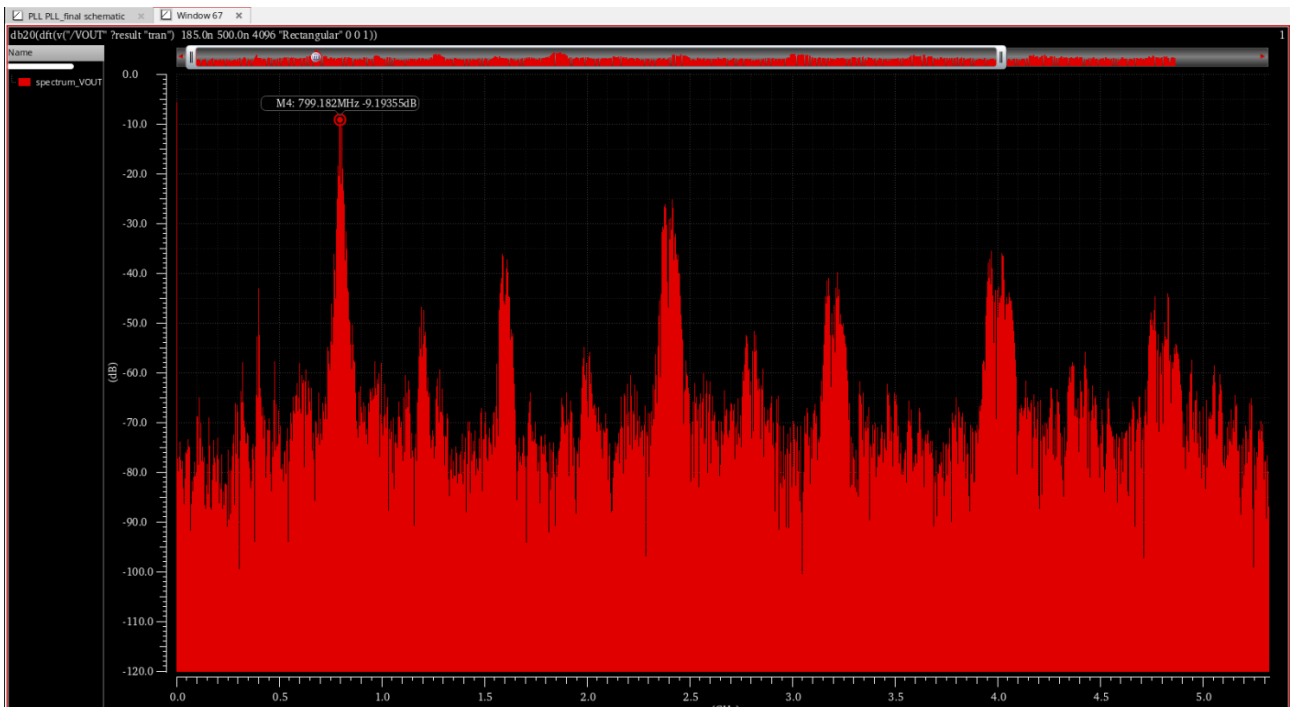


Figure 35 — FFT spectrum of a PLL Performance with frequency Divide-by-4 Configuration

#### 4.4.5 PLL Performance with frequency Divide-by-5 Configuration

##### Transient simulation

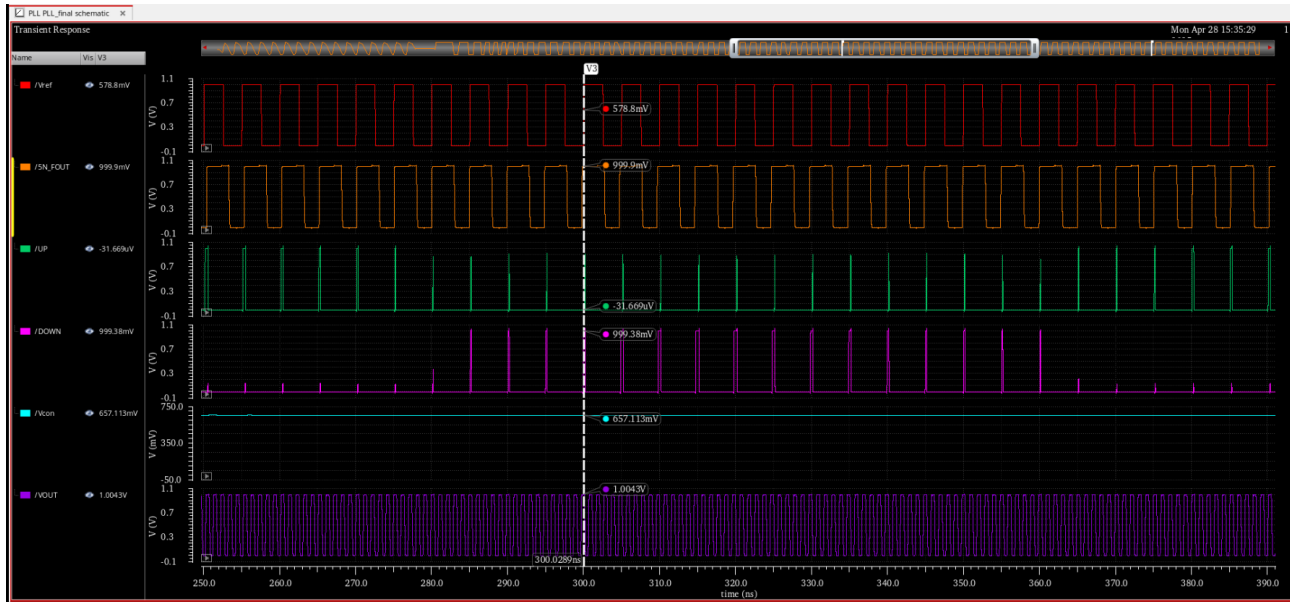


Figure 36 — Transient simulation of a PLL Performance with frequency Divide-by-5 Configuration

##### FFT spectrum

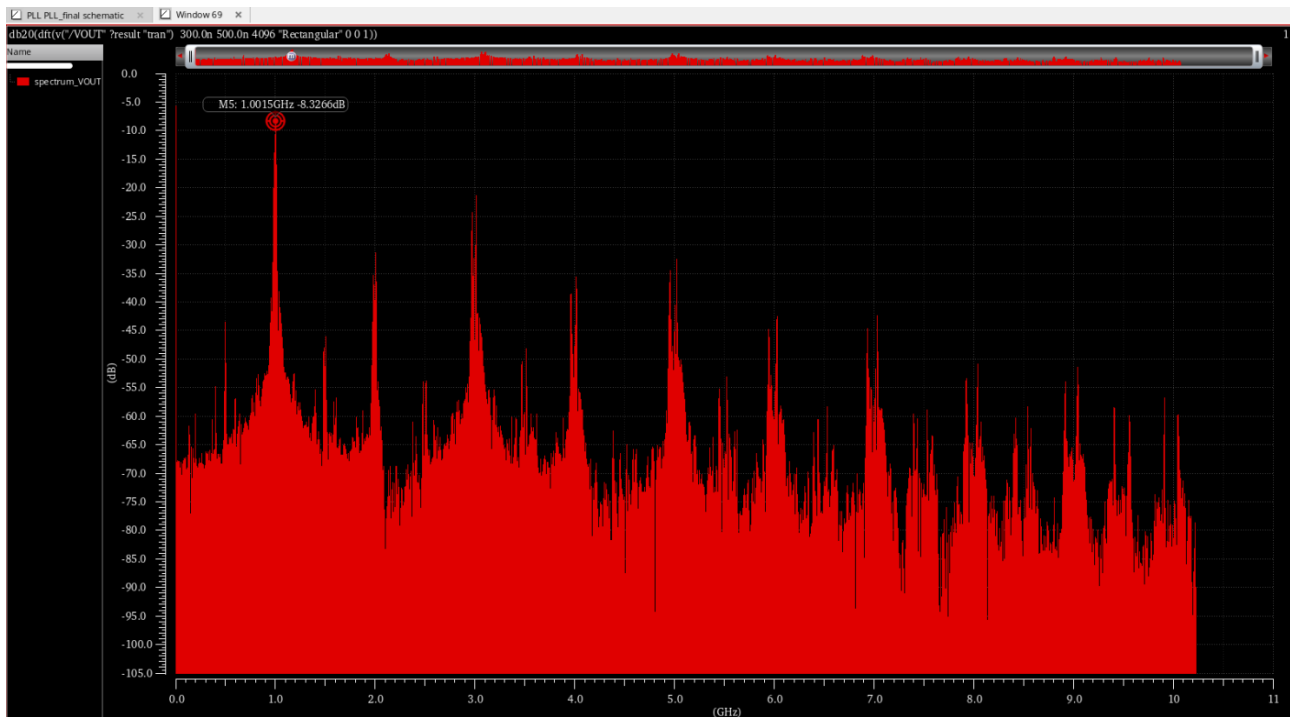


Figure 37 — FFT spectrum of a PLL Performance with frequency Divide-by-5 Configuration

#### 4.4.6 PLL Performance with frequency Divide-by-6 Configuration

##### Transient simulation

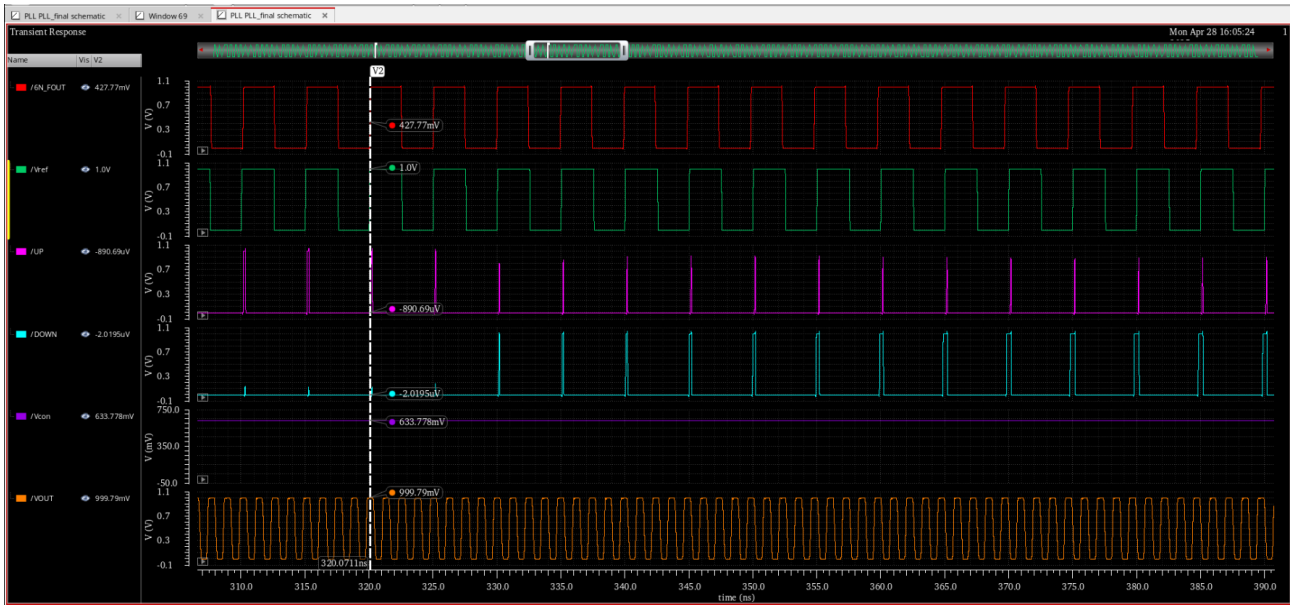


Figure 38 — Transient simulation of a PLL Performance with frequency Divide-by-6 Configuration

##### FFT spectrum

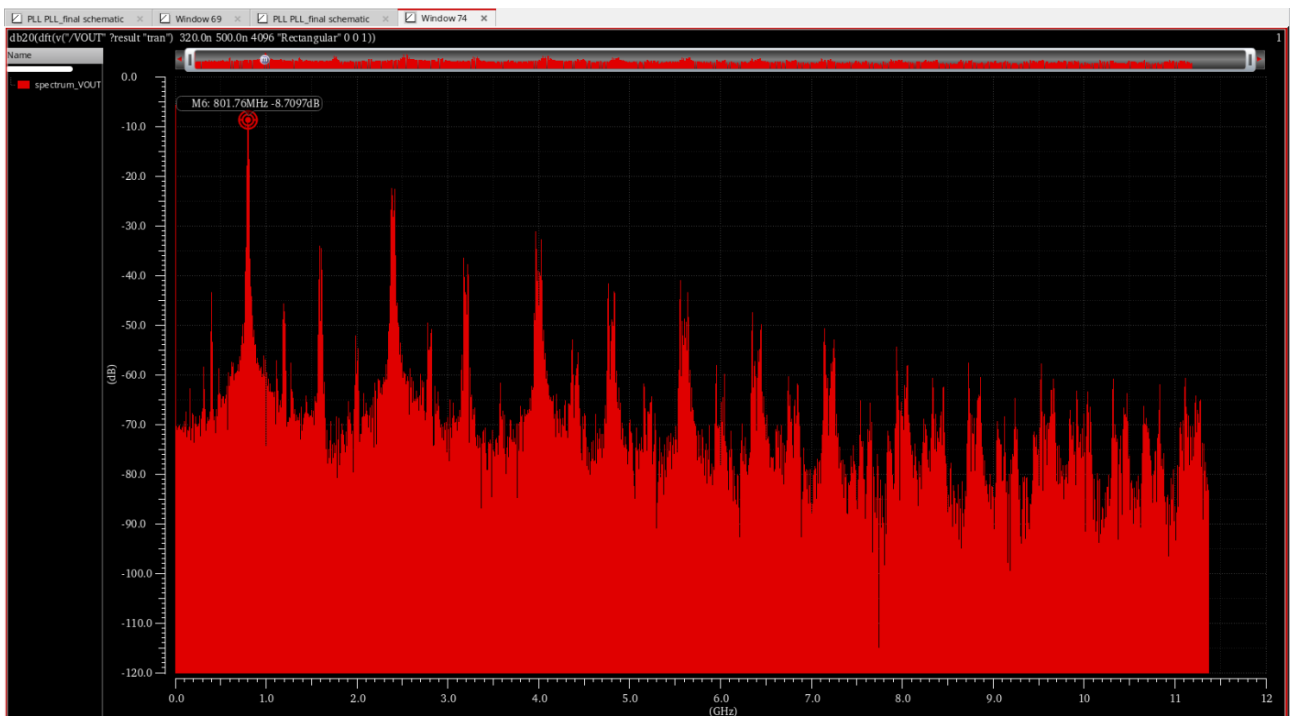


Figure 39 — FFT spectrum of a PLL Performance with frequency Divide-by-6 Configuration

#### 4.4.7 PLL Performance with frequency Divide-by-8 Configuration

##### Transient simulation

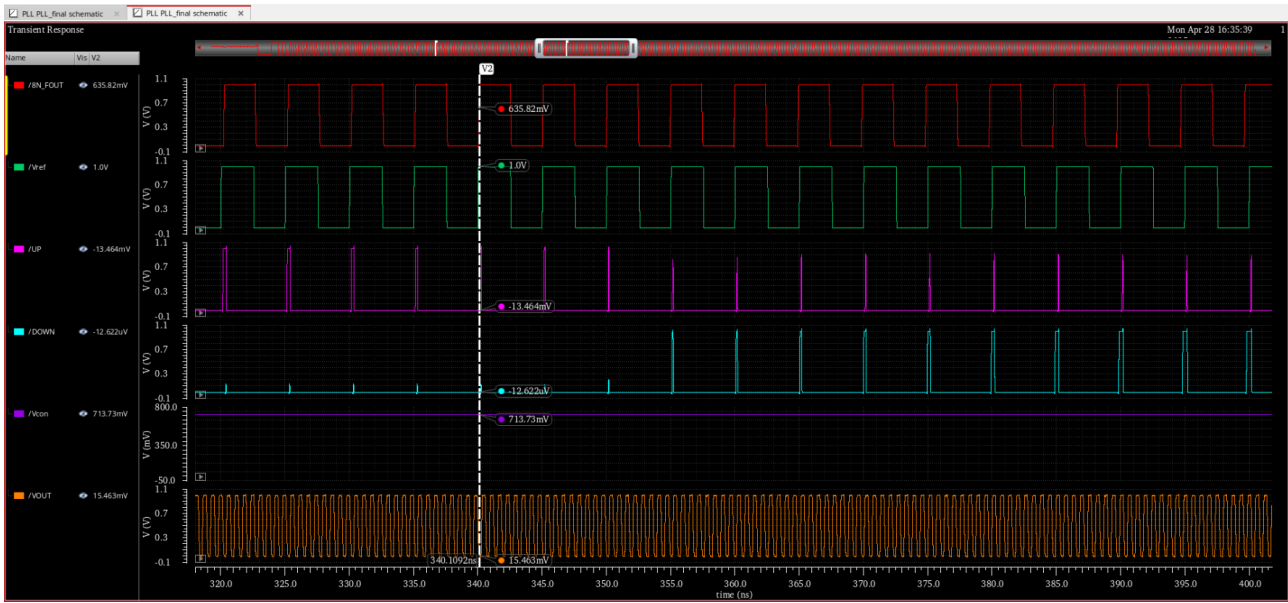


Figure 40 — Transient simulation of a PLL Performance with frequency Divide-by-8 Configuration

##### FFT spectrum

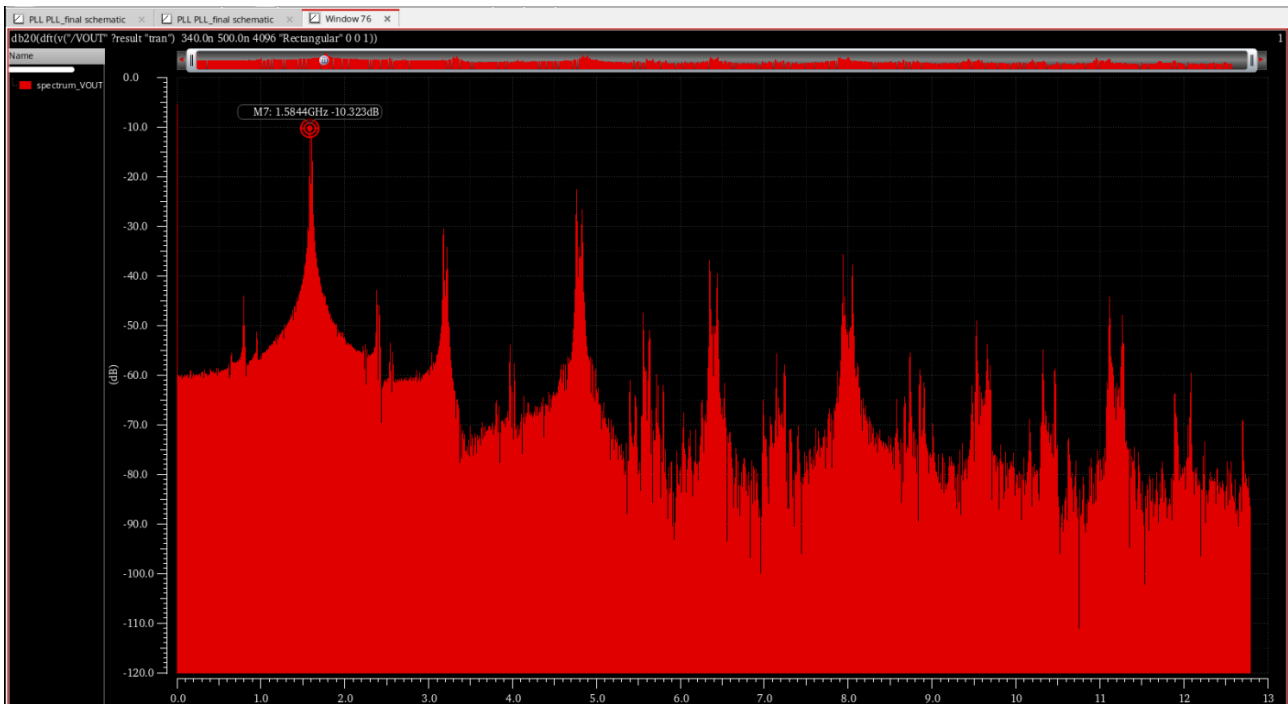


Figure 41 — FFT spectrum of a PLL Performance with frequency Divide-by-8 Configuration

#### 4.4.8 PLL Performance with frequency Divide-by-9 Configuration

##### Transient simulation

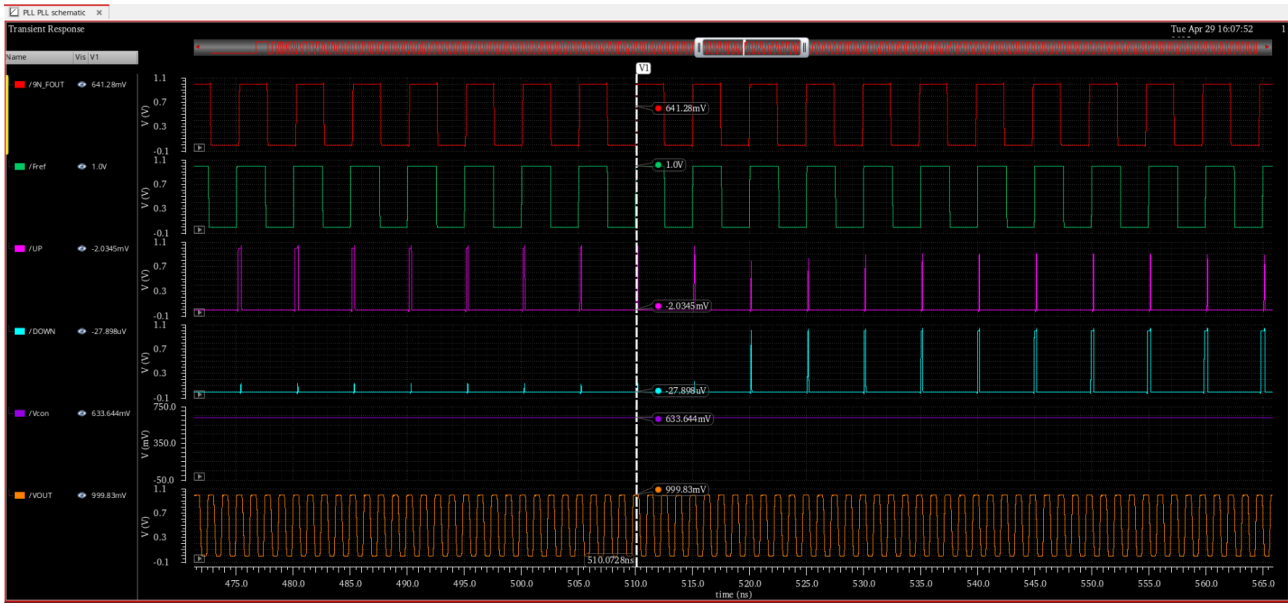


Figure 42 — Transient simulation of a PLL Performance with frequency Divide-by-9 Configuration

##### FFT spectrum

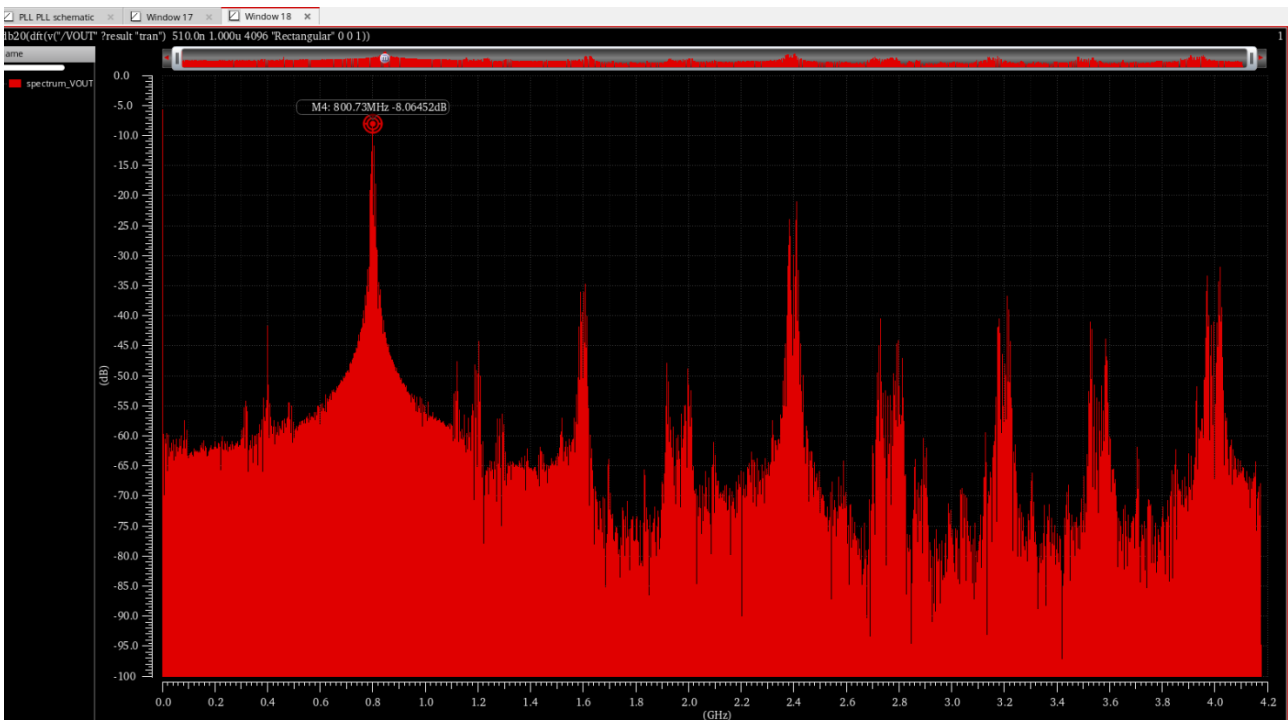


Figure 43 — FFT spectrum of a PLL Performance with frequency Divide-by-9 Configuration

#### 4.4.9 PLL Performance with frequency Divide-by-10 Configuration

##### Transient simulation

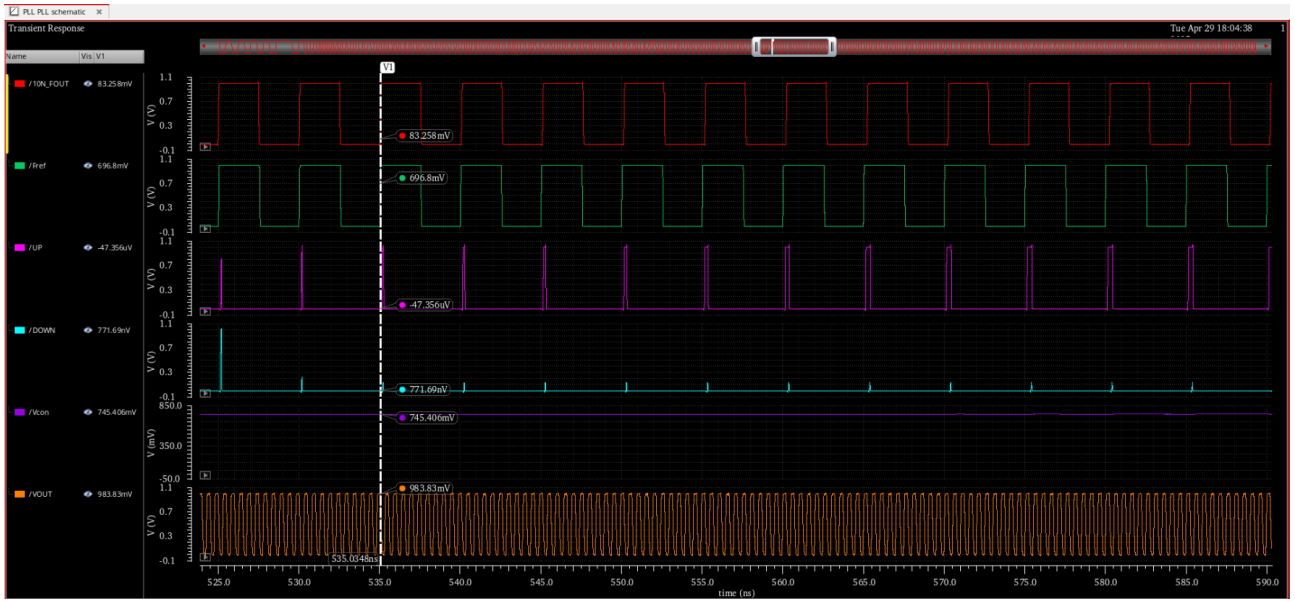


Figure 44 — Transient simulation of a PLL Performance with frequency Divide-by-10 Configuration

##### FFT spectrum

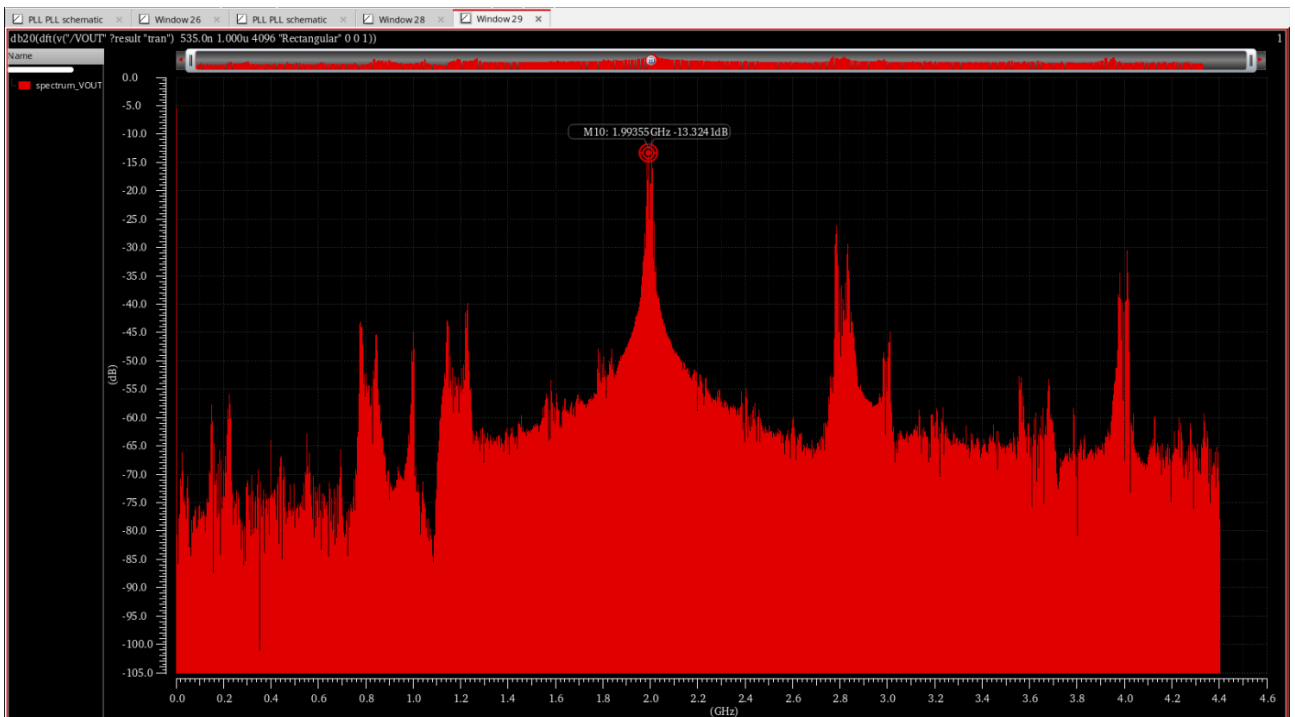


Figure 45 — FFT spectrum of a PLL Performance with frequency Divide-by-10 Configuration

#### 4.4.10 PLL Performance with frequency Divide-by-12 Configuration

##### Transient simulation

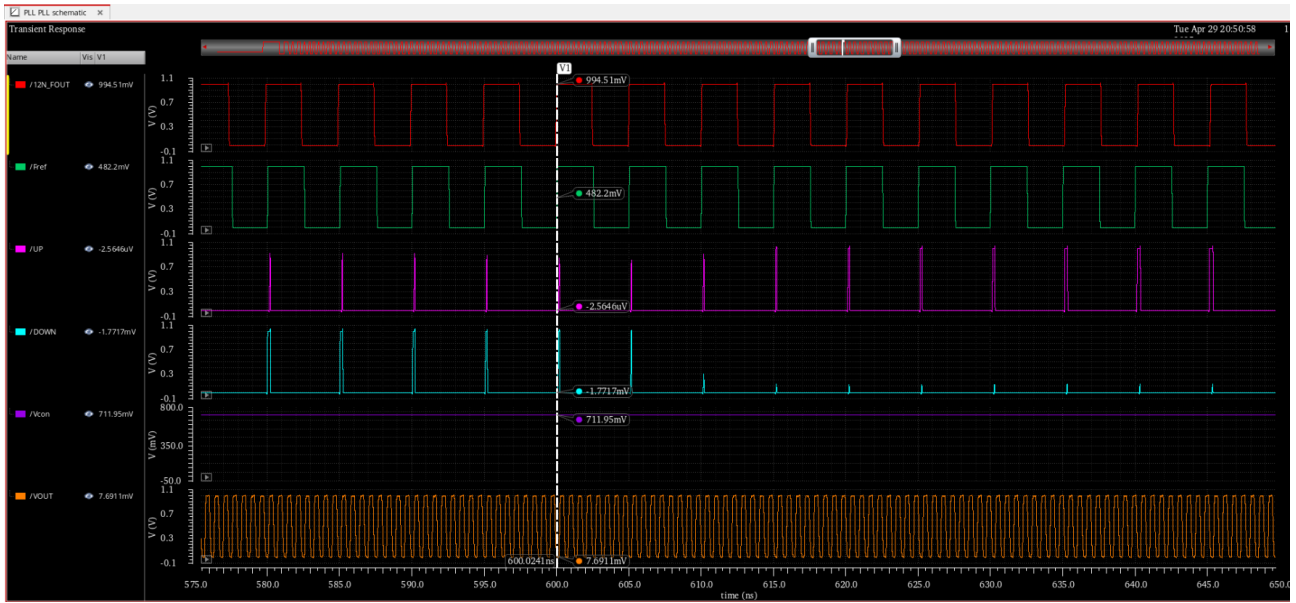


Figure 46 — Transient simulation of a PLL Performance with frequency Divide-by-12 Configuration

##### FFT spectrum

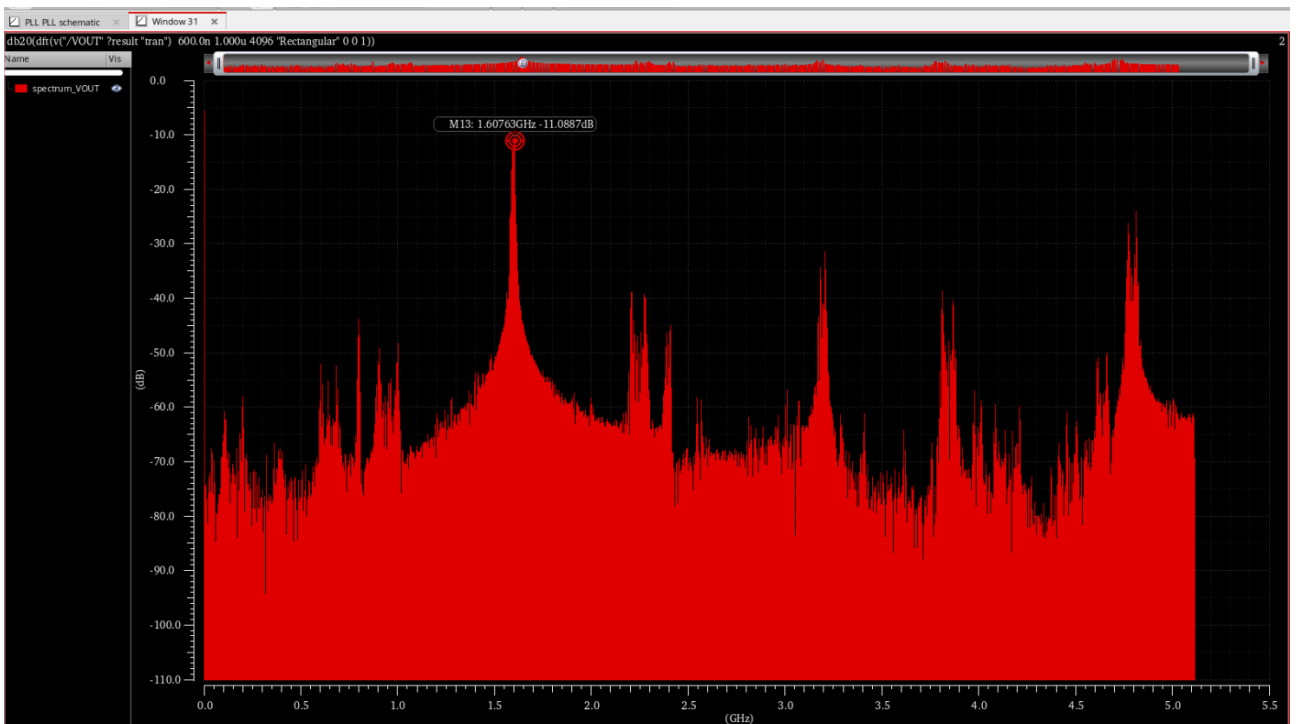


Figure 47 — FFT spectrum of a PLL Performance with frequency Divide-by-12 Configuration

#### 4.4.11 PLL Performance with frequency Divide-by-15 Configuration

##### Transient simulation

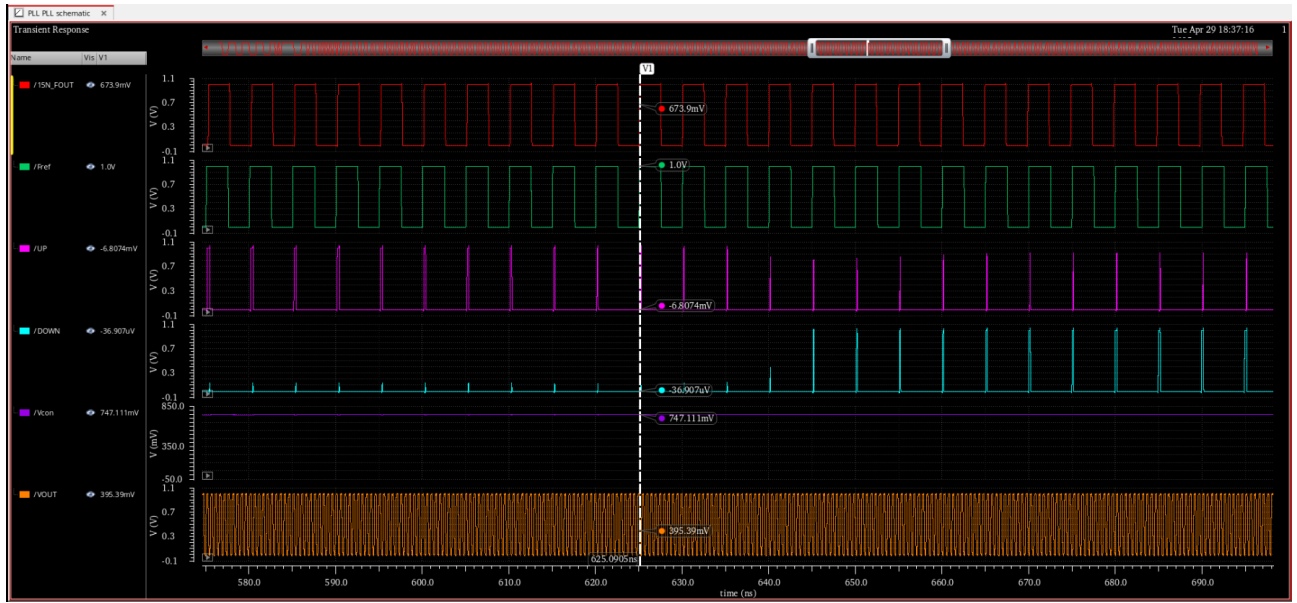


Figure 48 — Transient simulation of a PLL Performance with frequency Divide-by-15 Configuration

##### FFT spectrum

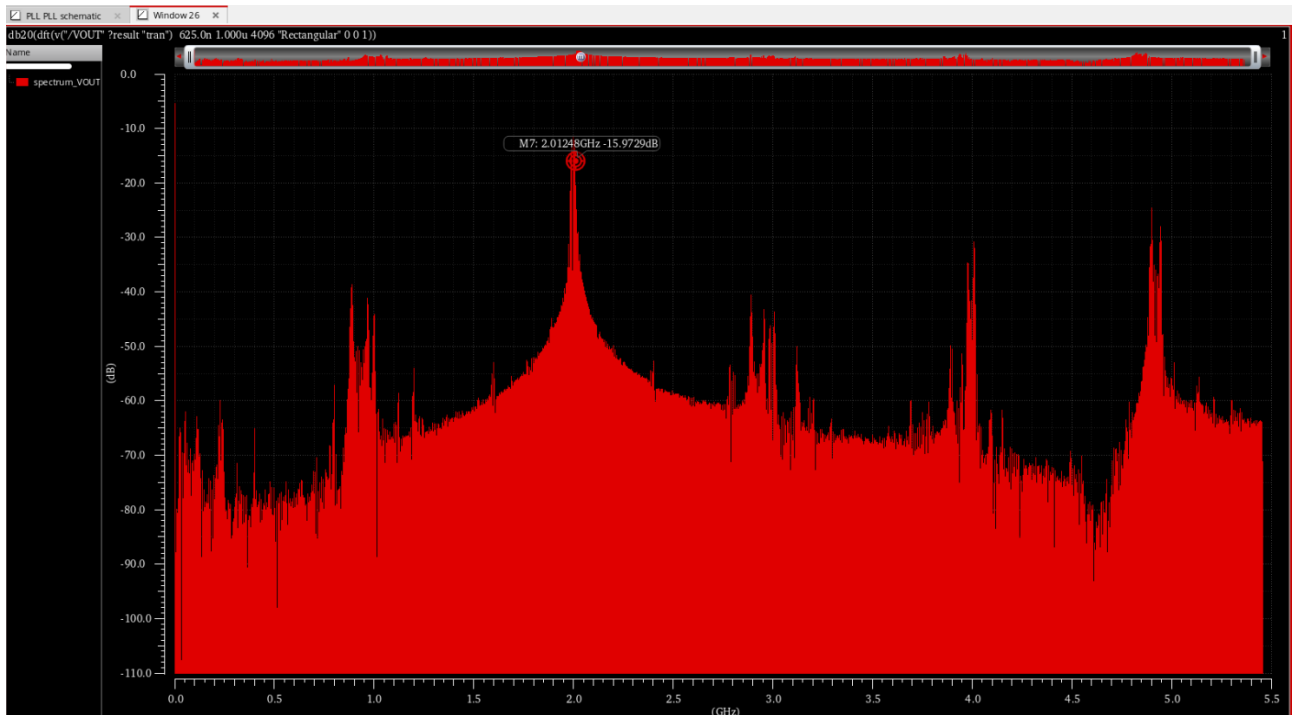


Figure 49 — FFT spectrum of a PLL Performance with frequency Divide-by-15 Configuration

#### 4.4.12 PLL Performance with frequency Divide-by-16 Configuration

##### Transient simulation

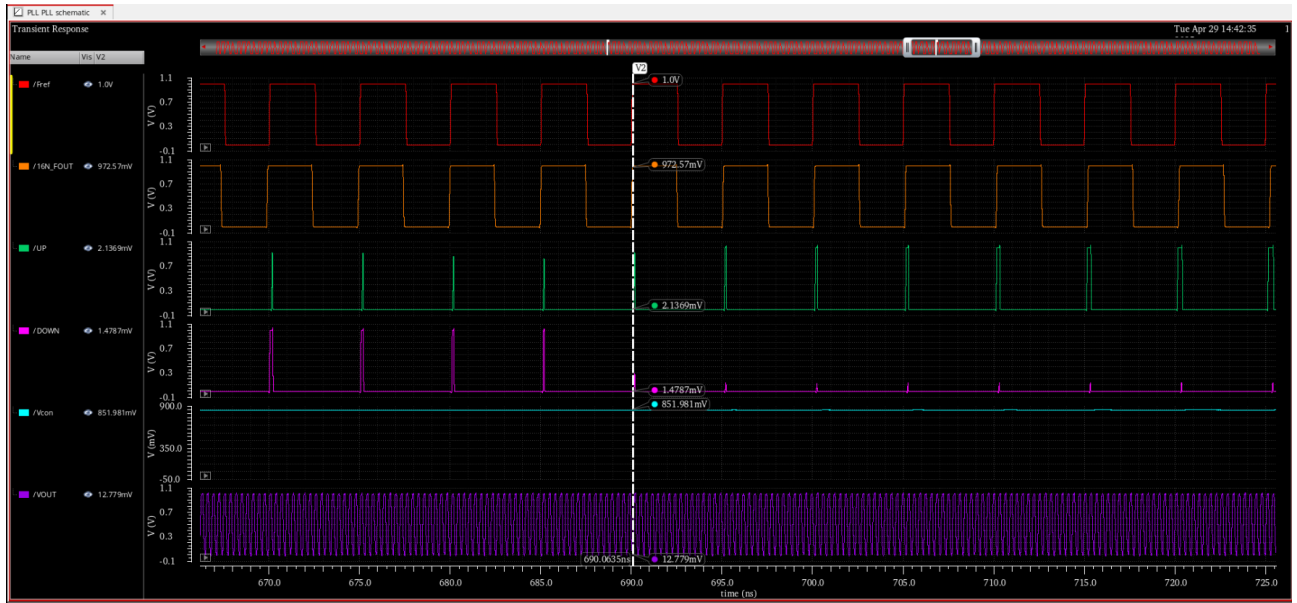


Figure 50 — Transient simulation of a PLL Performance with frequency Divide-by-16 Configuration

##### FFT spectrum

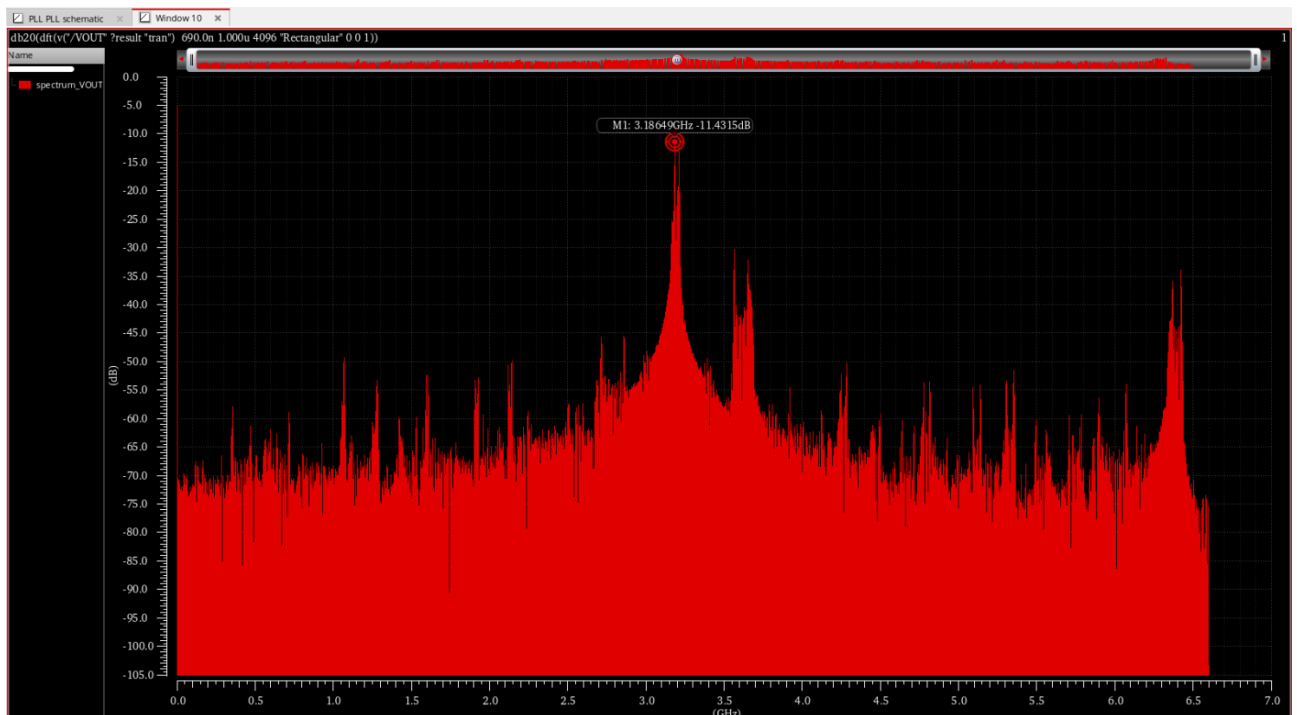


Figure 51 — FFT spectrum of a PLL Performance with frequency Divide-by-16 Configuration

## 5 SUMMARY OF OUTPUT FREQUENCY, LOCKING TIME, POWER CONSUMPTION, AND FURTHER IMPROVEMENTS A PFD-CP TYPE-II $\Delta\Sigma$ FRACTIONAL-N PHASE LOCKED LOOP SIMULATION

Performance data was collected for divider ratios  $N = 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 16$ . The output frequency ( $f_{\text{OUT}}$ ) of a 5-stage Ring VCO and the locking time for each ratio are given below:

| Divider Ratio (N) | Output Frequency $f_{\text{OUT}}$ (GHz) | Locking Time ( $\mu\text{s}$ ) |
|-------------------|-----------------------------------------|--------------------------------|
| 2                 | 0.398969                                | 0.145                          |
| 3                 | 0.399927                                | 0.16012                        |
| 4                 | 0.799182                                | 0.185                          |
| 5                 | 1.0015                                  | 0.300                          |
| 6                 | 0.80176                                 | 0.32                           |
| 8                 | 1.5844                                  | 0.3401                         |
| 9                 | 800.73                                  | 0.510                          |
| 10                | 1.99355                                 | 0.535                          |
| 12                | 1.60763                                 | 0.600                          |
| 15                | 2.01248                                 | 0.625                          |
| 16                | 3.1864                                  | 0.690                          |

Table 1 — Output Frequency and Locking Time for Various Divider Ratios

### 5.1 Output Frequency vs. Divider Ratio

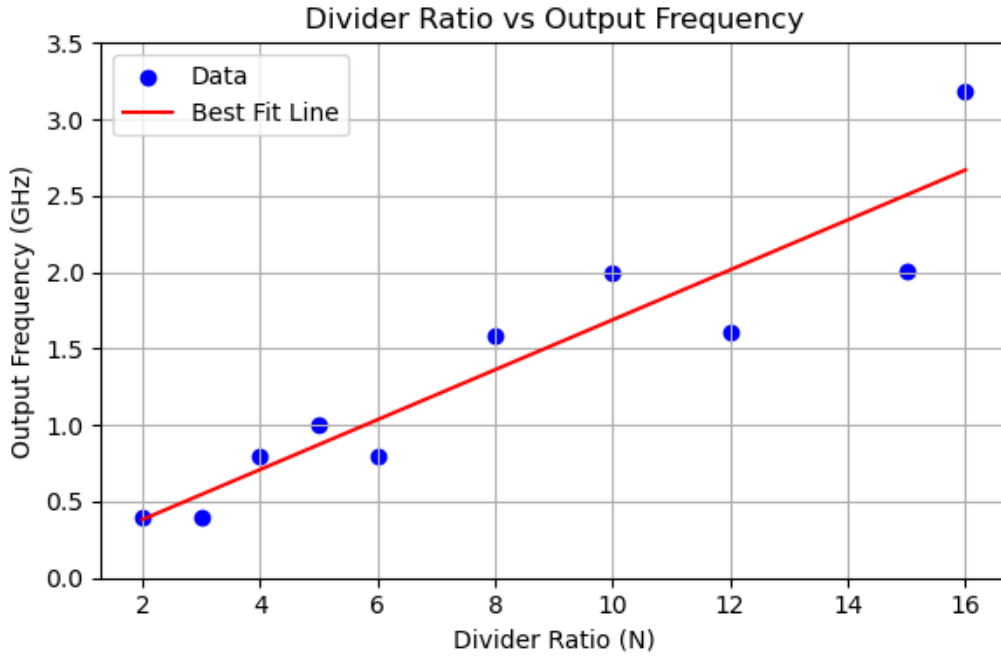


Figure 52 — The output frequency  $f_{\text{OUT}}$  Vs the divider ratio  $N$ .

The output frequency  $f_{\text{OUT}}$  increases with the divider ratio  $N$ .  
The range of output frequencies for  $2 \leq N \leq 16$  is:

$$f_{\text{OUT}} \in [0.3989 \text{ GHz}, 3.1864 \text{ GHz}]$$

The graph (Figure 51) of  $f_{\text{OUT}}$  vs.  $N$  shows a nearly linear relationship, meaning:

$$f_{\text{OUT}} \approx N \times f_{\text{REF}}$$

where  $f_{\text{REF}}$  is the reference frequency.

## 5.2 Locking Time vs. Divider Ratio

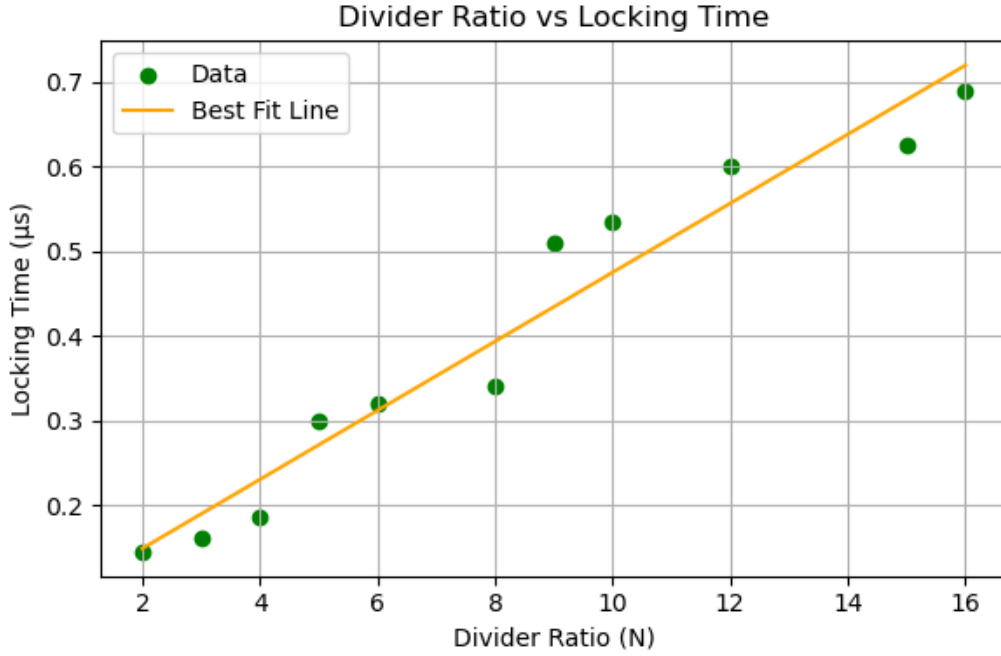


Figure 53 — The Locking time Vs the divider ratio  $N$ .

Locking time also increases as the divider ratio  $N$  increases.  
The range of locking times is:

$$\text{Locking Time} \in [0.145 \mu\text{s}, 0.690 \mu\text{s}]$$

This is shown in Figure 52, which plots locking time versus divider ratio.

## 5.3 Overall Power Consumption

The power usage of the Integer-N Phase-Locked Loop (PLL) was analyzed for a 1 V supply voltage, considering dynamic and static power.

### 5.3.1 Dynamic Power Consumption

Dynamic power comes from the switching of the circuit. It is calculated by:

$$P_{\text{dynamic}} = V \times I_{\text{avg}}$$

where  $V = 1 \text{ V}$  and  $I_{\text{avg}}$  is the average current.

For Divider Ratio  $N = 2$

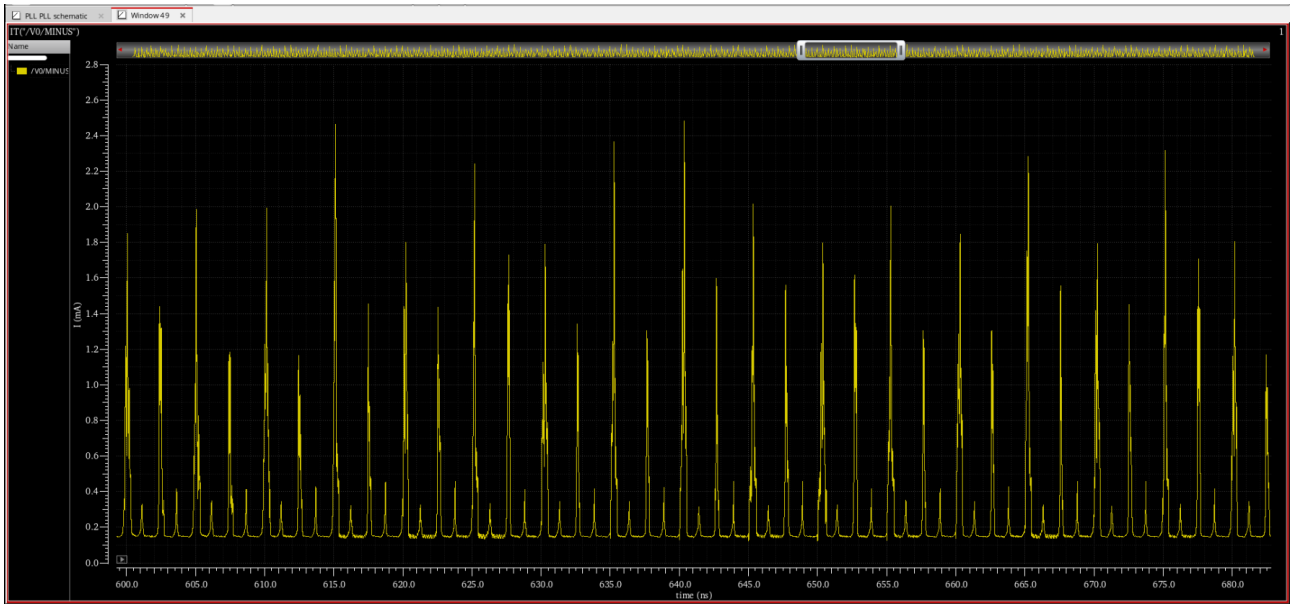


Figure 54 — Average current For Divider Ratio  $N = 2$

$$I_{\text{avg}} = 269.9 \mu A$$

$$P_{\text{dynamic}} = 1 V \times 269.9 \mu A = 269.9 \mu W$$

For Divider Ratio  $N = 16$

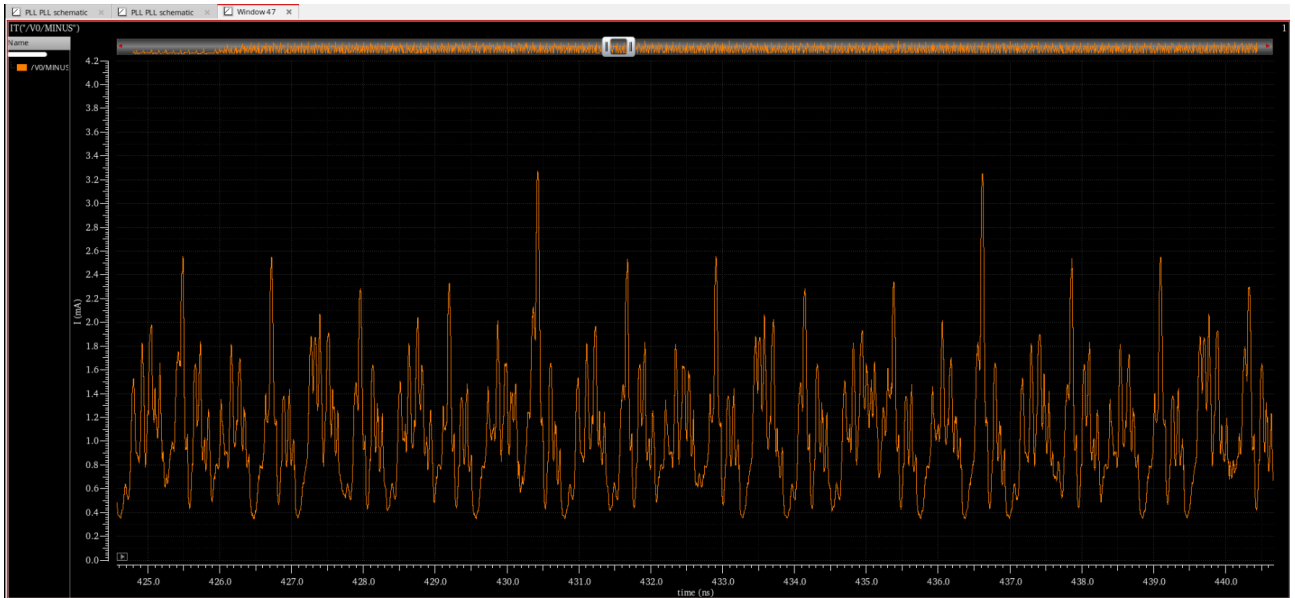


Figure 55 — Average current For Divider Ratio  $N = 2$

$$I_{\text{avg}} = 990 \mu A$$

$$P_{\text{dynamic}} = 1 V \times 990 \mu A = 990 \mu W$$

### 5.3.2 Static Power Consumption

There is no static power consumption because there are no extra DC sources other than the 1V supply.

### 5.3.3 Total Power Consumption

Since static power is zero, the total power equals the dynamic power. Therefore, for  $2 \leq N \leq 16$ , the total power consumption ranges from:

$$\boxed{269.9 \mu\text{W} \quad \text{to} \quad 990 \mu\text{W}}$$

### 5.4 Improvements for Fractional-N PLL Performance

1. **Lower Phase Noise:** Use a high-Q VCO ( $Q = 50\text{--}2000$ ) with low  $K_{\text{VCO}}$  (around  $2 \text{ MHz/V}$ ). A programmable loop filter can be used to set the bandwidth around  $f_{\text{REF}}/5$ . This can improve phase noise by  $10\text{--}20 \text{ dB}$ .
2. **Faster Lock Time:** Use a dual-mode loop filter—wide bandwidth during startup and narrow after locking. This is important for quick frequency changes, like in mobile networks.
3. **Better Power Efficiency:** Optimize the divider and PFD circuits in  $45\text{nm}$  CMOS technology, use power-gating techniques, and apply multi-loop systems like mix-and-count-down. These methods can reduce power consumption by  $20\text{--}30\%$ .

## 6 APPENDIX: NETLISTS

### 6.1 Netlist of a PFD-CP Type-II $\Delta\Sigma$ Fractional-N Phase Locked Loop

```
1
2 // Generated for: spectre
3 // Generated on: Apr 29 21:25:17 2025
4 // Design library name: PLL
5 // Design cell name: PLL
6 // Design view name: schematic
7 simulator lang=spectre
8 global 0
9 include "/home/aed/cadence/dicd_source/cadence_pdk/gpdk045_v_6_0/gpdk045/.../
   models/spectre/gpdk045.scs" section=mc
10
11 // Library name: PFD
12 // Cell name: NAND
13 // View name: schematic
14 subckt NAND A B VDD VOUT
15     PM1 (VOUT B VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
16         ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
17         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
18     PM0 (VOUT A VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
19         ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
20         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
21     NM1 (net16 A 0 0) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f ps=2.28u \
22         pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n sca=72.26287 \
23         scb=0.06569 scc=0.00861 m=(1)
24     NM0 (VOUT B net16 0) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
25         ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
26         sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
27 ends NAND
28 // End of subcircuit definition.
29
30 // Library name: PFD
31 // Cell name: NAND_3
32 // View name: schematic
33 subckt NAND_3 A B C VDD VOUT
34     PM2 (VOUT C VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
35         ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
36         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
37     PM1 (VOUT B VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
38         ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
39         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
40     PM0 (VOUT A VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
41         ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
42         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
43     NM2 (net8 C 0 net8) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
44         ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
45         sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
46     NM1 (net16 B net8 net8) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
47         ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
48         sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
49     NM0 (VOUT A net16 net8) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
50         ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
51         sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
52 ends NAND_3
53 // End of subcircuit definition.
54
55 // Library name: PFD
```

```

56 // Cell name: Inverter
57 // View name: schematic
58 subckt Inverter INPUT VDD VOUT VSS
59     NMO (VOUT INPUT VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
60         ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
61         sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
62     PMO (VOUT INPUT VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
63         ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
64         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
65 ends Inverter
66 // End of subcircuit definition.
67
68 // Library name: PFD
69 // Cell name: PFD_Final
70 // View name: schematic
71 subckt PFD_Final DOWN DOWNB Ffeed Fref UP UPB VDD VSS
72     I10 (net35 net34 VDD net40) NAND
73     I9 (net20 net20 VDD net34) NAND
74     I8 (net25 net16 VDD net20) NAND
75     I7 (net21 net21 VDD net35) NAND
76     I6 (net15 net18 VDD net21) NAND
77     I5 (DOWNB net36 VDD net16) NAND
78     I4 (net26 net16 VDD net25) NAND
79     I3 (net40 net25 VDD net26) NAND
80     I2 (net18 net40 VDD net28) NAND
81     I1 (net15 net28 VDD net18) NAND
82     I0 (UPB net39 VDD net15) NAND
83     I12 (net40 net25 net16 VDD DOWNB) NAND_3
84     I11 (net15 net18 net40 VDD UPB) NAND_3
85     I16 (UPB VDD UP VSS) Inverter
86     I15 (DOWNB VDD DOWN VSS) Inverter
87     I14 (Ffeed VDD net36 VSS) Inverter
88     I13 (Fref VDD net39 VSS) Inverter
89 ends PFD_Final
90 // End of subcircuit definition.
91
92 // Library name: charge_pump
93 // Cell name: Charge_pump_final
94 // View name: schematic
95 subckt Charge_pump_final DOWN DOWNB UP UPB VDD VSS Vcon
96     PM8 (net17 net17 VDD VDD) g45p1svt w=(4u) l=500n nf=1 as=560f ad=560f \
97         ps=8.28u pd=8.28u nrd=35m nrs=35m sa=140n sb=140n sd=160n \
98         sca=22.49589 scb=0.02034 scc=0.00174 m=(1)
99     PM7 (net13 net13 VDD VDD) g45p1svt w=(20u) l=120n nf=2 as=2.8p ad=1.6p \
100         ps=40.56u pd=20.32u nrd=4m nrs=7m sa=140n sb=140n sd=160n \
101         sca=34.56345 scb=0.04089 scc=0.00363 m=(1)
102     PM6 (net22 net13 VDD VDD) g45p1svt w=(20u) l=120n nf=2 as=2.8p ad=1.6p \
103         ps=40.56u pd=20.32u nrd=4m nrs=7m sa=140n sb=140n sd=160n \
104         sca=34.56345 scb=0.04089 scc=0.00363 m=(1)
105     PM5 (Vcon net17 net049 VDD) g45p1svt w=(11u) l=200n nf=2 as=1.54p \
106         ad=880f ps=22.56u pd=11.32u nrd=7.27273m nrs=12.7273m sa=140n \
107         sb=140n sd=160n sca=30.99520 scb=0.03464 scc=0.00282 m=(1)
108     PM4 (net049 net19 VDD VDD) g45p1svt w=(11u) l=200n nf=2 as=1.54p \
109         ad=880f ps=22.56u pd=11.32u nrd=7.27273m nrs=12.7273m sa=140n \
110         sb=140n sd=160n sca=30.99520 scb=0.03464 scc=0.00282 m=(1)
111     PM3 (net19 net17 net045 VDD) g45p1svt w=(11u) l=200n nf=2 as=1.54p \
112         ad=880f ps=22.56u pd=11.32u nrd=7.27273m nrs=12.7273m sa=140n \
113         sb=140n sd=160n sca=30.99520 scb=0.03464 scc=0.00282 m=(1)
114     PM2 (net045 net19 VDD VDD) g45p1svt w=(11u) l=200n nf=2 as=1.54p \
115         ad=880f ps=22.56u pd=11.32u nrd=7.27273m nrs=12.7273m sa=140n \
116         sb=140n sd=160n sca=30.99520 scb=0.03464 scc=0.00282 m=(1)
117     PM1 (net33 net33 VDD VDD) g45p1svt w=(500n) l=200n nf=1 as=70f ad=70f \
118         ps=1.28u pd=1.28u nrd=280m nrs=280m sa=140n sb=140n sd=160n \

```

```

119     sca=84.52381 scb=0.06559 scc=0.00884 m=(1)
120 PM0 (net19 net33 VDD VDD) g45p1svt w=(500n) l=200n nf=1 as=70f ad=70f \
121     ps=1.28u pd=1.28u nrd=280m nrs=280m sa=140n sb=140n sd=160n \
122     sca=84.52381 scb=0.06559 scc=0.00884 m=(1)
123 NM10 (net3 net22 VSS VSS) g45n1svt w=(10u) l=200n nf=1 as=1.4p ad=1.4p \
124     ps=20.28u pd=20.28u nrd=14m nrs=14m sa=140n sb=140n sd=160n \
125     sca=28.31345 scb=0.03320 scc=0.00255 m=(1)
126 NM9 (net43 net22 VSS VSS) g45n1svt w=(10u) l=200n nf=1 as=1.4p ad=1.4p \
127     ps=20.28u pd=20.28u nrd=14m nrs=14m sa=140n sb=140n sd=160n \
128     sca=28.31345 scb=0.03320 scc=0.00255 m=(1)
129 NM8 (Vcon net28 net43 VSS) g45n1svt w=(10u) l=200n nf=1 as=1.4p \
130     ad=1.4p ps=20.28u pd=20.28u nrd=14m nrs=14m sa=140n sb=140n \
131     sd=160n sca=28.31345 scb=0.03320 scc=0.00255 m=(1)
132 NM7 (net22 net28 net3 VSS) g45n1svt w=(10u) l=200n nf=1 as=1.4p \
133     ad=1.4p ps=20.28u pd=20.28u nrd=14m nrs=14m sa=140n sb=140n \
134     sd=160n sca=44.12978 scb=0.04989 scc=0.00563 m=(1)
135 NM6 (net10 net28 VSS VSS) g45n1svt w=(2u) l=200n nf=1 as=280f ad=280f \
136     ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
137     sca=41.18123 scb=0.04022 scc=0.00387 m=(1)
138 NM5 (net28 net28 VSS VSS) g45n1svt w=(2u) l=200n nf=1 as=280f ad=280f \
139     ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
140     sca=41.18123 scb=0.04022 scc=0.00387 m=(1)
141 NM4 (net6 net28 VSS VSS) g45n1svt w=(2u) l=200n nf=1 as=280f ad=280f \
142     ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
143     sca=41.18123 scb=0.04022 scc=0.00387 m=(1)
144 NM3 (net22 DOWNB net10 VSS) g45n1svt w=(2u) l=120n nf=1 as=280f \
145     ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
146     sca=47.43123 scb=0.04792 scc=0.00496 m=(1)
147 NM2 (net13 DOWN net10 VSS) g45n1svt w=(2u) l=120n nf=1 as=280f ad=280f \
148     ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
149     sca=47.43123 scb=0.04792 scc=0.00496 m=(1)
150 NM1 (net19 UP net6 VSS) g45n1svt w=(2u) l=120n nf=1 as=280f ad=280f \
151     ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
152     sca=47.43123 scb=0.04792 scc=0.00496 m=(1)
153 NM0 (net33 UPB net6 VSS) g45n1svt w=(2u) l=120n nf=1 as=280f ad=280f \
154     ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
155     sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
156 C1 (Vcon VSS) capacitor c=1p
157 C0 (net044 VSS) capacitor c=5p
158 R0 (Vcon net044) resistor r=1K
159 I6 (net17 net28) isource dc=80u type=dc
160 ends Charge_pump_final
161 // End of subcircuit definition.
162
163 // Library name: Ring_VCO
164 // Cell name: Ring_VCO_final
165 // View name: schematic
166 subckt Ring_VCO_final VDD VOUT VSS mirror_ng
167     I5 (Feedback net30 net8 net27) Inverter
168     I4 (net8 net29 net9 net26) Inverter
169     I3 (Feedback VDD VOUT VSS) Inverter
170     I2 (net11 net22 Feedback net21) Inverter
171     I1 (net10 net23 net11 net24) Inverter
172     I0 (net9 net28 net10 net25) Inverter
173     PM6 (mirror_pg mirror_pg VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f \
174         ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
175         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
176     PM4 (net22 mirror_pg VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f \
177         ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
178         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
179     PM3 (net23 mirror_pg VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f \
180         ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
181         sca=56.99756 scb=0.05691 scc=0.00695 m=(1)

```

```

182 PM2 (net28 mirror_pg VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f \
183 ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
184 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
185 PM1 (net29 mirror_pg VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f \
186 ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
187 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
188 PM0 (net30 mirror_pg VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f \
189 ad=280f ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
190 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
191 NM5 (mirror_pg mirror_ng VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f \
192 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \
193 sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
194 NM4 (net21 mirror_ng VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f \
195 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \
196 sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
197 NM3 (net24 mirror_ng VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f \
198 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \
199 sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
200 NM2 (net25 mirror_ng VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f \
201 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \
202 sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
203 NM1 (net26 mirror_ng VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f \
204 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \
205 sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
206 NM0 (net27 mirror_ng VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f \
207 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \
208 sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
209 ends Ring_VCO_final
210 // End of subcircuit definition.
211
212 // Library name: PFD
213 // Cell name: DFF
214 // View name: schematic
215 subckt DFF CLK D Q Q_bar VDD VSS
216 PM4 (net13 CLK net28 VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
217 ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
218 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
219 PM3 (Q Q_bar VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
220 ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
221 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
222 PM2 (Q_bar net16 VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
223 ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
224 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
225 PM1 (net16 CLK VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
226 ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
227 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
228 PM0 (net28 D VDD VDD) g45p1svt w=(2u) l=45n nf=1 as=280f ad=280f \
229 ps=4.28u pd=4.28u nrd=70m nrs=70m sa=140n sb=140n sd=160n \
230 sca=56.99756 scb=0.05691 scc=0.00695 m=(1)
231 NM5 (Q Q_bar VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
232 ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
233 sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
234 NM4 (net26 net16 VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
235 ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
236 sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
237 NM3 (net27 CLK VSS net23) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
238 ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
239 sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
240 NM2 (net13 D VSS VSS) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
241 ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
242 sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
243 NM1 (net16 net13 net27 net23) g45n1svt w=(1u) l=45n nf=1 as=140f \
244 ad=140f ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n \

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245         sd=160n sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
246     NMO (Q_bar CLK net26 VSS) g45n1svt w=(1u) l=45n nf=1 as=140f ad=140f \
247         ps=2.28u pd=2.28u nrd=140m nrs=140m sa=140n sb=140n sd=160n \
248         sca=72.26287 scb=0.06569 scc=0.00861 m=(1)
249     C1 (Q VSS) capacitor c=10f
250     C0 (Q_bar VSS) capacitor c=10f
251 ends DFF
252 // End of subcircuit definition.
253
254 // Library name: freq_divider
255 // Cell name: freq_divider_3N
256 // View name: schematic
257 subckt freq_divider_3N CLK_IN CLK_OUT Q1_bar VDD VSS
258     I1 (net14 net010 VDD net19) NAND
259     I3 (net19 VDD net20 VSS) Inverter
260     I5 (CLK_IN net20 CLK_OUT net14 VDD VSS) DFF
261     I4 (CLK_IN net14 Q1_bar net010 VDD VSS) DFF
262 ends freq_divider_3N
263 // End of subcircuit definition.
264
265 // Library name: freq_divider
266 // Cell name: freq_divider_5N
267 // View name: schematic
268 subckt freq_divider_5N CLK_IN CLK_OUT Q1 VDD VSS
269     I1 (net14 net16 VDD net19) NAND
270     I0 (net12 M VDD net18) NAND
271     I3 (net19 VDD net20 VSS) Inverter
272     I2 (net18 VDD net17 VSS) Inverter
273     I6 (net14 CLK_OUT M CLK_OUT VDD VSS) DFF
274     I5 (CLK_IN net20 net12 net14 VDD VSS) DFF
275     I4 (CLK_IN net17 Q1 net16 VDD VSS) DFF
276 ends freq_divider_5N
277 // End of subcircuit definition.
278
279 // Library name: freq_divider
280 // Cell name: multiple_freq_divider
281 // View name: schematic
282 subckt multiple_freq_divider \10N_FOUT \12N_FOUT \15N_FOUT \16N_FOUT \
283     \2N_FOUT \3N_FOUT \4N_FOUT \5N_FOUT \6N_FOUT \8N_FOUT \9N_FOUT \
284     CLK_IN Q11 Q12 Q13 Q1_bar1 Q1_bar2 Q1_bar3 Q1_bar4 VDD VSS
285     I13 (\6N_FOUT Q1_bar4 \12N_FOUT Q1_bar4 VDD VSS) DFF
286     I12 (\5N_FOUT Q12 \10N_FOUT Q12 VDD VSS) DFF
287     I3 (\8N_FOUT net39 \16N_FOUT net39 VDD VSS) DFF
288     I2 (\4N_FOUT net36 \8N_FOUT net36 VDD VSS) DFF
289     I1 (\2N_FOUT net33 \4N_FOUT net33 VDD VSS) DFF
290     I0 (CLK_IN net43 \2N_FOUT net43 VDD VSS) DFF
291     I15 (\5N_FOUT \15N_FOUT Q13 VDD VSS) freq_divider_3N
292     I6 (\2N_FOUT \6N_FOUT Q1_bar3 VDD VSS) freq_divider_3N
293     I5 (\3N_FOUT \9N_FOUT Q1_bar2 VDD VSS) freq_divider_3N
294     I4 (CLK_IN \3N_FOUT Q1_bar1 VDD VSS) freq_divider_3N
295     I8 (CLK_IN \5N_FOUT Q11 VDD VSS) freq_divider_5N
296 ends multiple_freq_divider
297 // End of subcircuit definition.
298
299 // Library name: PLL
300 // Cell name: PLL
301 // View name: schematic
302 I0 (DOWN DOWNB \16N_FOUT Fref UP UPB VDD 0) PFD_Final
303 I1 (DOWN DOWNB UP UPB VDD 0 Vcon) Charge_pump_final
304 I2 (VDD VOUT 0 Vcon) Ring_VCO_final
305 I3 (\10N_FOUT \12N_FOUT \15N_FOUT \16N_FOUT \2N_FOUT \3N_FOUT \4N_FOUT \
306     \5N_FOUT \6N_FOUT \8N_FOUT \9N_FOUT VOUT Q11 Q12 Q13 Q1_bar1 \
307     Q1_bar2 Q1_bar3 Q1_bar4 VDD 0) multiple_freq_divider

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```
308 V0 (VDD 0) vsource dc=1 type=dc
309 V1 (Fref 0) vsource type=pulse val0=0 val1=1 period=5n rise=50p fall=50p \
310     width=2.5n
311 ic VOUT=1
312 simulatorOptions options reltol=1e-3 vabstol=1e-6 iabstol=1e-12 temp=27 \
313     tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 maxnotes=5 maxwarns=5 \
314     digits=5 cols=80 pivrel=1e-3 sensfile="../psf/sens.output" \
315     checklimitdest=psf
316 tran tran stop=1u errpreset=moderate write="spectre.ic" \
317     writefinal="spectre.fc" annotate=status maxiters=5
318 finalTimeOP info what=oppoint where=rawfile
319 modelParameter info what=models where=rawfile
320 element info what=inst where=rawfile
321 outputParameter info what=output where=rawfile
322 designParamVals info what=parameters where=rawfile
323 primitives info what=primitives where=rawfile
324 subckts info what=subckts where=rawfile
325 save V0:p V0:p V1:p V1:p
326 saveOptions options save=allpub
```

Listing 6.1 — Cadence-generated Netlist